

User Manual

Breakdown Management System

Revision: R0

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PREFACE

The user manual provides detailed description of application System's functionality (Breakdown Management System), which help any user to understand the system.

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1 INTRODUCTION

The propose solution comprehensive login management with robust authentication and authorization features, allowing users to securely access specific modules based on their roles. It integrates preventive maintenance (PM) and breakdown management (BM), optimizing resource allocation and ensuring equipment reliability. User management is streamlined, with admins assigning roles and permissions to various user types, facilitating efficient workflow within departments. The BM request sheet dashboard enables easy generation and tracking of maintenance requests, while the approval dashboard streamlines the approval process, enhancing decision-making. Real-time monitoring and advanced filtration options provide actionable insights, contributing to operational efficiency. Additionally, features like KPI reports, target setting, and data migration ensure comprehensive performance analysis and continuous improvement. Email functionality and other features like spare part management further enhance communication and operational effectiveness.

2 SYSTEM ARCHITECTURE

There are eight types of users in the system:

1. Admin
2. MTD HOD
3. PRD HOD
4. MTD HOS
5. PRD HOS
6. MTD TL/HOSS
7. PRD TL/HOSS
8. Operator

Following table gives the user permissions to access the various pages:

3 SOFTWARE DESCRIPTION

3.1 Overview

The web application has the following screens/pages:

- Login
- BM Tab
 1. Main dashboards
 - a) Request-sheet progress monitoring
 - b) Request-sheet status monitoring
 2. Approval dashboard
 3. Approval logs
 4. Master logs
 5. Other loss
 6. Customized dashboard

- MTD KPI Tab

1. MTD KPI

2. Reports

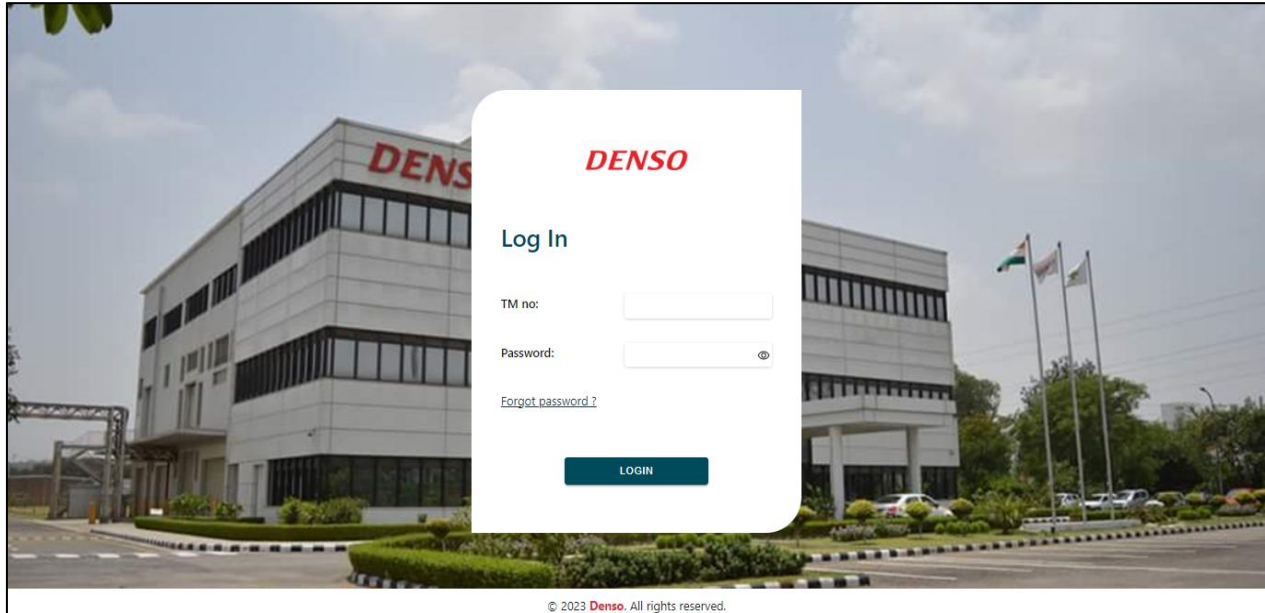
- a) Production line wise
- b) Man Hours
- c) Daily BD
- d) Monthly BD
- e) Line Contribution BD
- f) MTTR
- g) MTBF
- h) TM MTTR skill
- i) Top machine breakdown
- j) Machine age

3. Target dashboard

4. Spare reports

3.2 Web Application

3.2.1 Main Login



- The GUI shown above is the main login screen used to validate a user for access of dashboard. The user needs to enter the TM no.(Team Member No) and password and then click on “Login” button. If the credentials are correct, the user will be redirected to main dashboard.
- If a user forgets their password, they should enter their registered email address to receive a link to change their password.

3.2.2 Request-sheet progress monitoring

Sr. No.	Request No.	Product	Line	Machine No.	Machine Name	Problem	Date-time	R.S Status	Assign	Handover To	Final Action	H/O Time Work End	First Time/Repeat	Loss Time	W.O. Status	Actions
1	EL-MA2-5-20	Mounting	MA2	EETP-009	Board Flipper	hsh	02-05-2024 T09:16	Under MTD HOSS Approval	Jatinder Singh		Asdfqwer DFG dfg	02-05-2024 T10:16	Repeat	60	Closed	
2	EL-MA2-4-19	Mounting	MA2	EETP-044	SPI	QWERFG	24-04-2024 T16:10	Under MTD HOD Approval	Jatinder Singh		Oyuir xdvSDFDcfv fdfs WSERFG	24-04-2024 T18:33	First Time	143	Closed	
3	EL-MA2-4-18	Mounting	MA2	EETP-044	SPI	QWERFG	24-04-2024 T16:10	Generated							Open	
4	EL-MA2-4-17	Mounting	MA2	EETP-044	SPI	PCB transport error	24-04-2024 T16:01	Generated							Open	
5	EL-MA2-4-16	Mounting	MA2	EETP-094	SPG-2			Generated							Open	

- This dashboard showcases request sheets that have been generated, assigned, or completed within each approval stage.
- Users can access the overall count of generated requests, encompassing both open and closed statuses.
- Filtering options enable users to sort request sheets by cell/product, line, machine, and status. They can further refine their search by viewing request sheets by month or year. Additionally, users can retrieve request sheets by entering a value equal to or greater than the request sheet's loss time.

DENSO Integrated Maintenance Operation System

PM BM MTD KPI

Request Sheet Progress Monitoring GENERATE REQUEST SHEET SUMMARY

Total Requests: 136 Open Requests: 57 Closed Requests: 79

Select Cell 2024-2025 Month Req. Sheet Stat... RESET >> GO

Sl. No.	Request No.	Product	Line	Machine No.	Machine Name	Problem	Date-time	R.S. Status	Loss Time	W.O. Status	Actions
1	EL-MA2-5-20	Mounting	MA2	EETP-009	Board Flipper	hsh	02-05-2024 T09:16	Under MTD HOSS Approval	60	Closed	
2	EL-MA2-4-19	Mounting	MA2	EETP-044	SPI	QWERFG	24-04-2024 T16:10	Under MTD HOD Approval	143	Closed	
3	EL-MA2-4-18	Mounting	MA2	EETP-044	SPI	QWERFG	24-04-2024 T16:10	Generated		Open	
4	EL-MA2-4-17	Mounting	MA2	EETP-044	SPI	PCB transport error	24-04-2024 T16:01	Generated		Open	
5	EL-MA2-4-16	Mounting	MA2	EETP-094	SPG-2			Generated		Open	
6	EL-FA-WP3-4-22	Final Assy	FA-WP3	EEAT-156	Silicon All-round	pokayoke not working during daily check	24-04-2024 T07:19	Under MTD TL Approval	40	Closed	
7	EL-AOI-6-4-15	Mounting	AOI-6	EETP-089	AOI-06	Slow processing	23-04-2024 T15:55	Assigned		Open	
8	EL-MA5-4-7	Mounting	MA5	EETP-073	YS Loader	moving lifter time out error	23-04-2024 T18:00	Under MTD TL Approval	25	Closed	
9	EL-MA6-4-8	Mounting	MA6	EETP-115	BOARD FLIPPER + CLEANER	card jaam problem	23-04-2024 T17:10	Under MTD TL Approval	20	Closed	

- Users can toggle the visibility of columns using a blue toggle button, allowing them to display only the necessary information.
- All request sheet data can be downloaded in CSV format.
- After all approvals, if a user needs to edit certain fields that are inaccurate, they can do so by clicking on the edit button. Access to this button is restricted to users assigned by their higher authorities on the "Assign user" dashboard under the PM's tab.
- Users can also delete or withdraw request sheets, with access to this function granted by higher authorities after approval, similar to editing request sheets.

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PM BM MTD KPI

User Assign Dashboard

12	149	Shrijoy
13	3430	Masaru Bess
14	3485	Tushar
15	1178	Sunil Poonia
16	2967	Ashish Gupta
17	89	Chanchal
18	843	Rajeev Gupta
19	954	Gaurav Sharma
20	3419	Vishnu
21	3431	Sandeep Dha

Edit User

TM Name: Ashish Gupta

TM Number: 2967

Want to authorized this user to update RequestSheet: Previously selected: Yes
☐ Yes ☐ No

Previous Tm Department: MTD
Previous User Type: TL/HOSS

TM Department: ☐ PRD ☐ MTD

Plant: P1-DNHA-P1

Section: S2-Electronics

Sub Section: Previous Sub-section: S54-ECU/AB, S55-Meter, S56-RTS

Cell/Product: Previous Cell: C1-Mounting, C2-Final Assy, C3-Meter Assy

Joining Date: Previous Date: 2015-08-03

Add PDF CSV

TS Assy, ...

1-Mounting

1-Mounting, C2-Final Assy, C3-Meter Assy

1-Mounting, C2-Final Assy, C3-Meter Assy

2-Final Assy

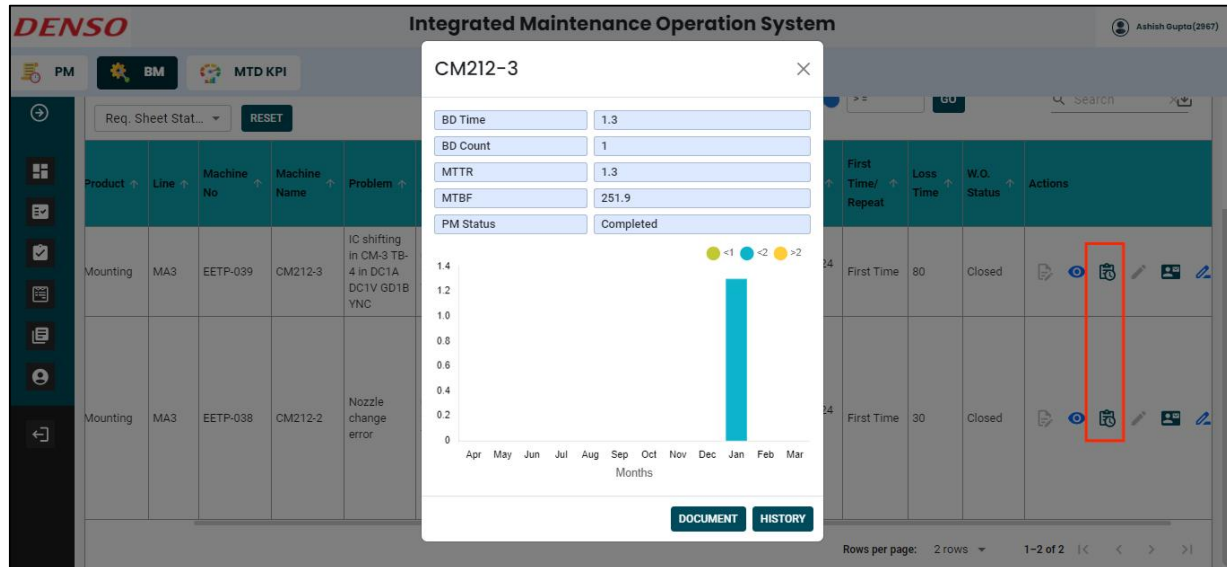
1-Mounting, C2-Final Assy, C3-Meter Assy

2-Final Assy

2-Final Assy

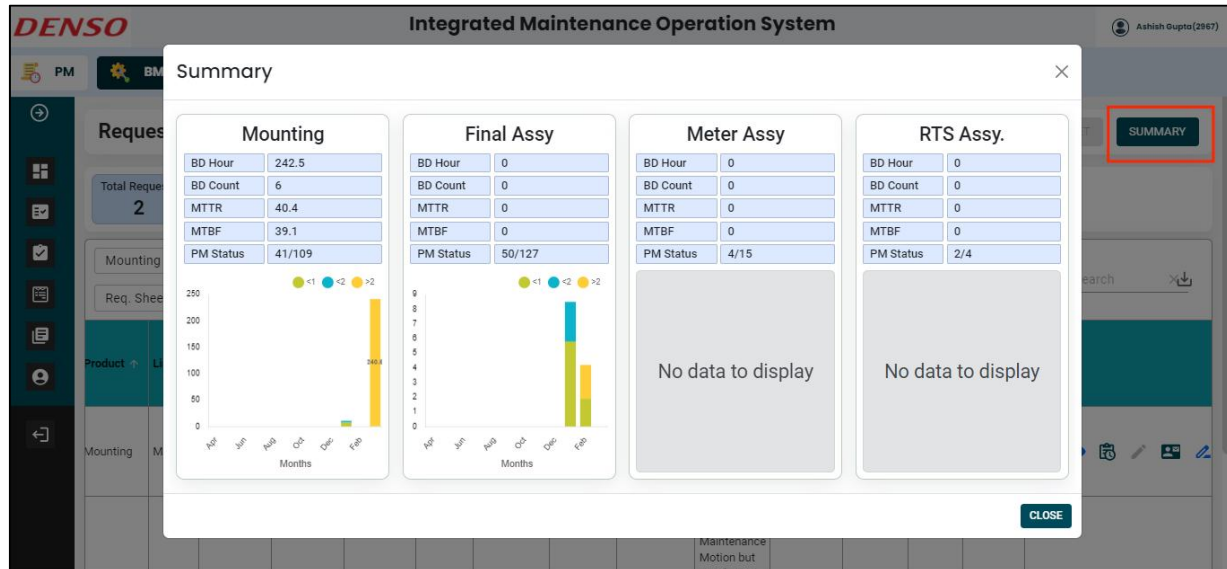
1-Mounting, C2-Final Assy, C3-Meter Assy

3.2.2.1 History card



- The machine history card features a chart displaying breakdown (BD) hours versus months, alongside metrics such as BD hours, Mean Time To Repair (MTTR), Mean Time Between Failures (MTBF), and PM status.
- Users can access documents pertaining to a specific machine, providing a comprehensive history including breakdown maintenance (BM) and preventive maintenance (PM) data.
- **Note:** To ensure accurate data representation, users must first set targets for each cell/product and line. Without user-defined targets, data may not be displayed correctly. It's essential to establish targets for both cell/product-wise and line-wise metrics.

3.2.2.2 Summary card



- This cell/product summary card have BD hours wise vs month chart and BD hours, MTTR, MTBF, PM plan vs actual count.
- Note: If target is not set by user the data not show properly. Firstly, need to set all target of cell/product wise and line wise.

3.2.2.3 Spare request form

The screenshot shows the 'Spare parts request' modal window in the DENSO Integrated Maintenance Operation System. It includes fields for Line name, Machine code, and Machine name. Below these fields is a table for Spare parts with columns: PART NO., PART NAME, MAKER, QUANTITY, Cost, and UPDATE. A 'Submit' button is at the bottom.

PART NO.	PART NAME	MAKER	QUANTITY	Cost	UPDATE

- On click of Spare require mail button one form will be open and user can fill the require spare list and submit.
- To save time, users can send spare requests to the registered tool room person's email address before reaching out.

3.2.2.4 Request-sheet view

Breakdown Request-Sheet Close

Machine Details PROBLEM HISTORY

MAINTENANCE WORK REQUEST/REPORT

PM Status: Current Plan | BM: 0.2 Hrs./1 Count | CM:

MAINT. TYPE BM	PRIORITY CODE EMERGENCY	REQUEST SHEET (To be filled by PRD) REQUEST No.EL-AOI-08-4-4 PROBLEM OCCURRED DATE & TIME: REQUEST SHEET (To be filled by PRD) REQUEST No.EL-AOI-08-4-4 PROBLEM OCCURRED DATE & TIME: 22-04-2024 09:10 AM	SHEET ISSUED DATE & TIME: SHEET ISSUED DATE & TIME: 22-04-2024 10:51	DEPT./LINE Mounting/AOI-08 TL [PRD] Parth Arjariya DEPT./LINE Mounting/AOI-08 TL [PRD] Parth Arjariya
BM	EMERGENCY			

MACHINE NAME: MULTIPLE MAGAZINE UNLOADER MACHINE NO.: EETP-133

PROBLEM FACED: Stocker width NG made by Machine

PRD OBSERVATION FOR THIS PROBLEM BASED ON (SWHY-IHOW): Observed when card stuck between invert

WHY (SM+IE): No changepoint.

WHERE (Process): Unloader

WHEN (Frequency): 100

WHO (Person): Rosiamliana

SHIFT: ☒ A ☐ B ☐ C

QUALITY RELATED: Yes

BREAKDOWN ATTENDED BY: Jatinder Singh

Breakdown Request-Sheet Close

MAINTENANCE REPORT (To be filled by MTD)

REQUEST RECEIVED: MTD S.L. Ashish Gupta | MTD TL: Sandeep Dhama

WORK STARTED	DATE & TIME: 22-04-2024 09:10 AM	WORK ENDED	DATE & TIME: 22-04-2024 09:20 AM	SECTION IN-CHARGE MTD HOSS Anshul Gupta
WORK STARTED	DATE & TIME: 22-04-2024 09:10 AM	WORK ENDED	DATE & TIME: 22-04-2024 09:20 AM	SECTION IN-CHARGE MTD HOSS Anshul Gupta

PROBLEM
Problem 1: PCB Jam Error Coming

WHY WHY ANALYSIS [ROOT CAUSE]

WHY-1
PCB Jam Error Coming

WHY-2
PCB Not Pass In Stocker

WHY-3
Stocker Width NG Made By Machine

WHY-4
Suspected Change Over Not Completed

WHY-5
NA

BREAKDOWN 10	ANALYSIS 5	SPARE WAITING 0	REPLACEMENT 0
	ADJUSTMENT 5	QUALITY CHECK 0	BREAK 0

MAJOR B/D: No
MINOR B/D: Yes

First Time Repeat

QUALITY CONFIRMED (IPP): Yes

**PART QUALITY CHECKED

PRD: Parth Arjariya | MTD: Sandeep Dhama

DATA SHEET ATTACHED: No

DRAWING ATTACHED: No

- User can view each request-sheet on click on eye icon.
- Also, user is able to download request-sheet as PDF format.

- User can also view problem history which are for the particular problem and machine.

Breakdown Request-Sheet

Problem Occurred: pokayoke not working during daily check

Machine Details **PROBLEM HIST**

MAINT. TYPE: BM

PRIORITY CC: EMERGENCY

MACHINE NAME: Silicon All-round

PROBLEM FACED: pokayoke

PRD OBSERVATION FOR THIS PROBLEM BASED ON (5WHY-THOW): no

WHY (5M+1E): no

WHERE (Process): si all around

WHEN (Frequency): 100

WHO (Person): vicky

WHICH (Defect Location): SI ALL AROUND

Sr. No.	Product	Line	Machine No	Machine Name	Request No	Final Action	Action and Counter Measure	Date-Time	Loss Time
1	Final Assy	FA-WP3	EEAT-156	Silicon All-round	EL-FA-WP3-4-22			24-04-2024 T07:19	40
2	Final Assy	FA-WP3	EEAT-156	Silicon All-round	EL-FA-WP3-4-19			22-04-2024 T09:30	10
3	Final Assy	FA-WP3	EEAT-156	Silicon All-round	EL-FA-WP3-4-8			08-04-2024 T20:48	12

Rows per page: 5 rows 1-3 of 3

DEPT./LINE: Final Assy/FA-WP3

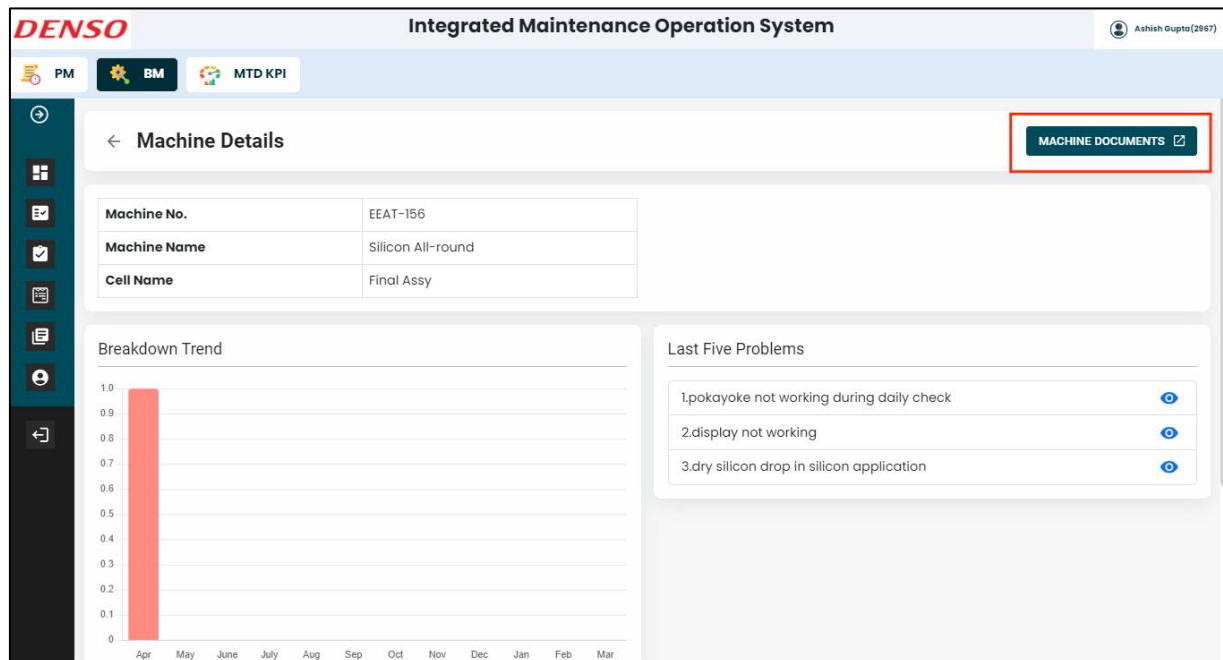
TL [PRD]: Shamsheel

DEED BY: talip Khajuria

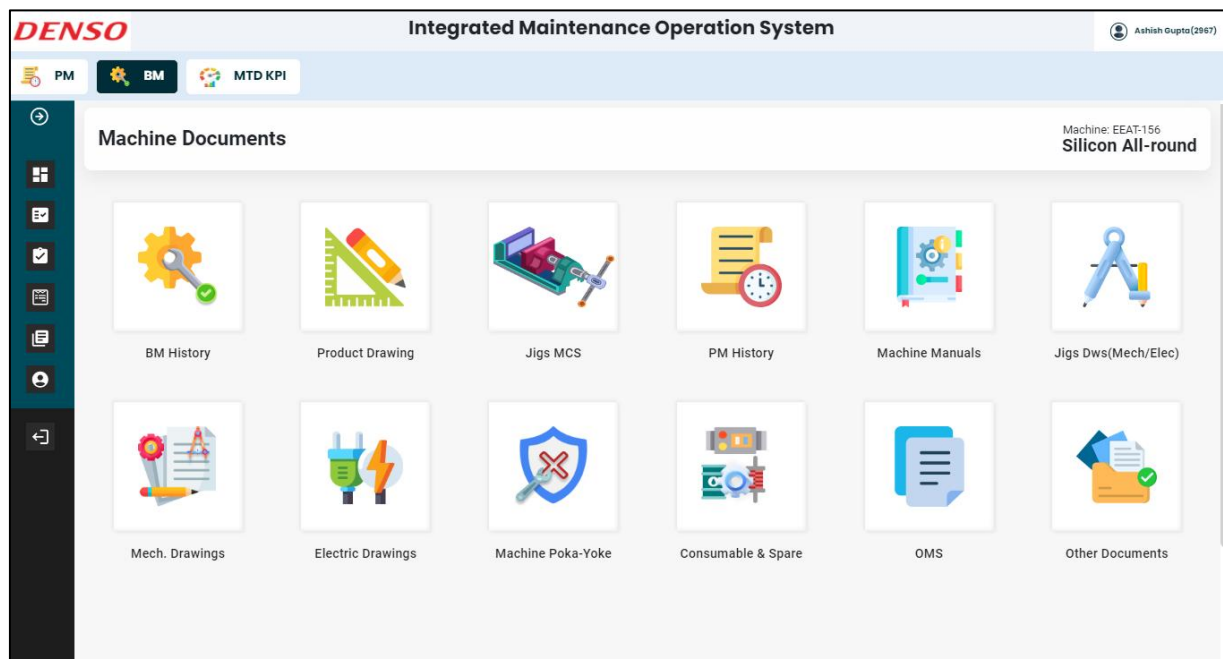
Close

3.2.2.5 Machine details and documents

- After viewing the BM request-sheet user can view machine details.
- In this details user is able to view breakdown trend and last five problem of that selected machine.



- On this screen there is one machine documents button, on click of the button below machine document screen is open.



- User can upload new document as per given categories and view/ delete the documents.
- Also user is able to view the history of BM and PM.

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Machine: EETP-009
Board Flipper

← BM History

2024-2025 Month

Sr. No.	Section	Request No.	Product	Line	Machine No.	Machine Name	Problem	Date-Time	Loss Time	Work Order Status	R.S Status	Actions
1	Electronics	EL-MA2-S-20	Mounting	MA2	EETP-009	Board Flipper	hsh	02-05-2024 T09:16	60	Closed	Under MTD HOSS Approval	

Rows per page: 5 rows 1-1 of 1

DENSO Integrated Maintenance Operation System

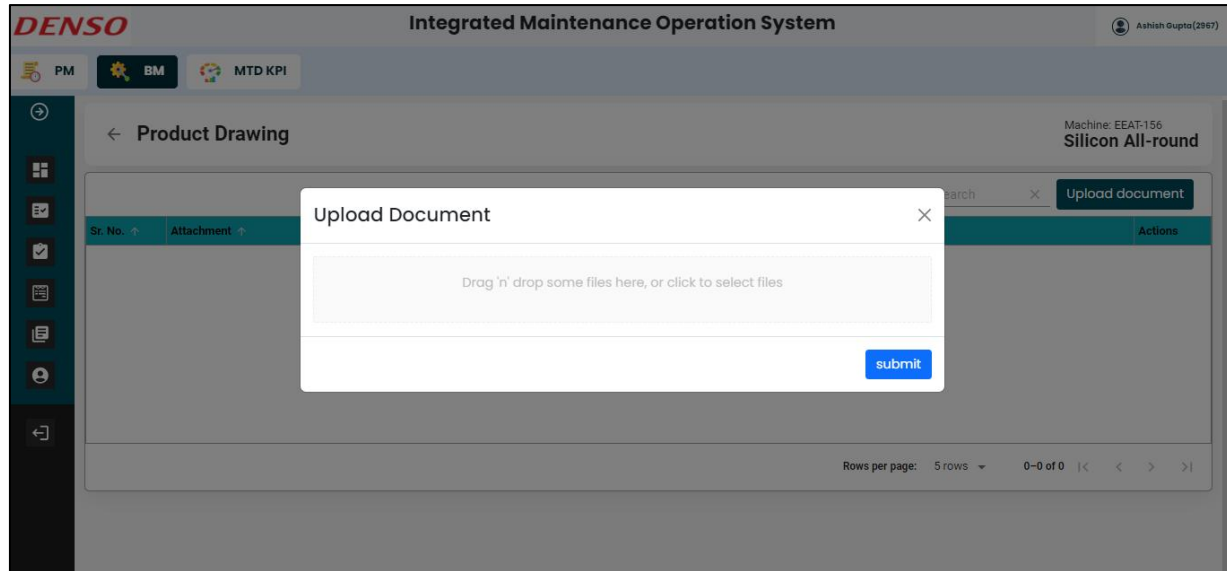
Machine: EEAT-156
Silicon All-round

← Product Drawing

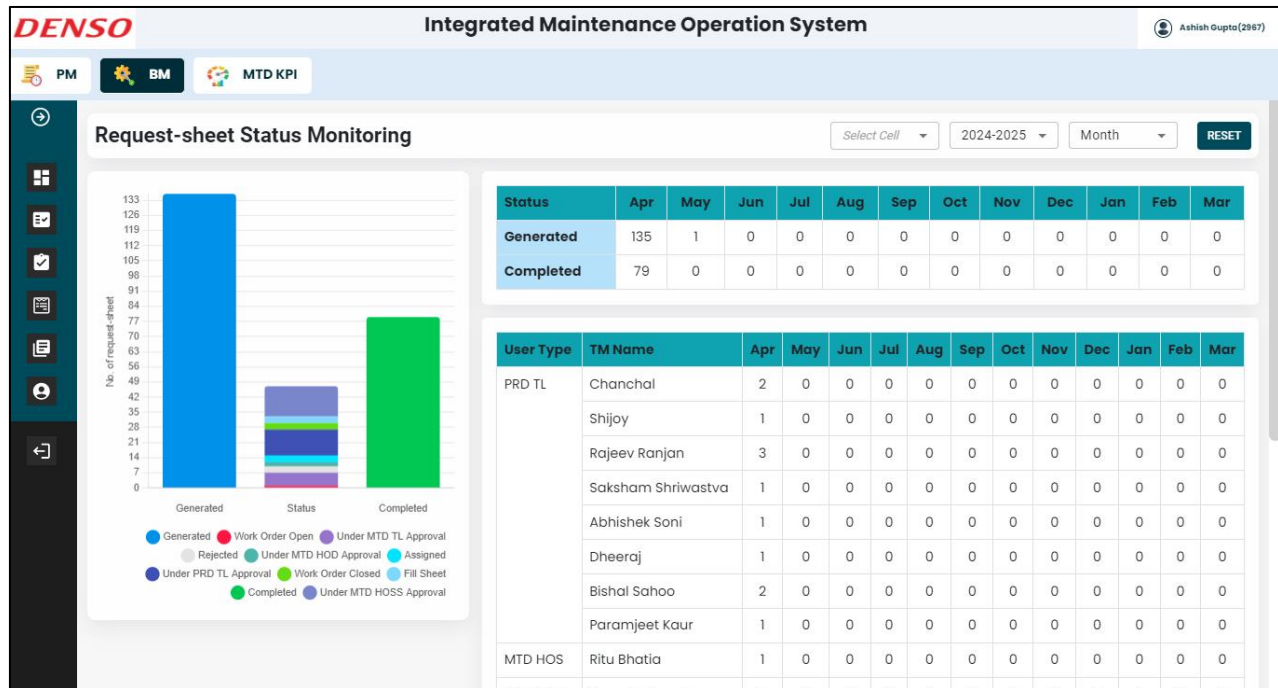
Search Upload document

Sr. No.	Attachment	Preview	Actions
No records to display			

Rows per page: 5 rows 0-0 of 0



3.2.3 Request-sheet status monitoring



- Request-sheet status monitoring dashboard contain all status wise chart with total generated, all status and completed vs no. of request-sheet count. So user is easily check-it out that how many request-sheet is persist to the particular status.
- Also it is display generated count and completed count with all months of the year.
- In this dashboard there is one user wise pending list display with month wise, based on this user is able take decision for it's approval approach and get his/her latest update.
- The user can filter request sheets counts based on cell/product, line and year/months wise.

3.2.4 Approval dashboard

- Whole approval flow is depend on the approval list which is selected by MTD HOD user.
- The approval flow is:
 1. After filling maintenance part by assign TM's, the TM's need to send approval to MTD TL.
 2. MTD TL review the request-sheet filled by assigned TM's and based on the minor or major break-down, MTD TL send approval to the require higher approval.
 - a) If MTD TL approve request-sheet and select all require higher authority (based on minor/major BD) the status of request-sheet is change from "Under MTD TL approval" to the "Under {next department grade} approval".
 - b) If reject, then MTD TL need to add rejected remarks and TM's will receive rejected request-sheet in "Request-sheet process monitoring dashboard" and again update the request-sheet as remarks and send approval to the MTD TL. The request-sheet status is on "Rejected".
 - c) MTD user is update the request-sheet while approve the request-sheet.
 3. The approval chain will continue up-to the last approval which is on approval list.
 - a) After MTD TL approve, If any of the higher authority will reject the request-sheet, then the MTD TL will receive rejected request-sheet in "Approval dashboard" and again update the request-sheet as remarks and send approval to the higher authority. The request-sheet status is on "Rejected". The assigned TM's not received rejected request-sheet by higher authority.
 4. The request-sheet status is change based on the next higher authority.
 5. While approval on MTD HOS stage need to add feedback by MTD HOS.
 6. After all approval of higher authority the request-sheet status on "Completed" stage.

DENSO Integrated Maintenance Operation System Ashish Gupta(2987)

PM BM MTD KPI

Approval Dashboard Select Cell 2024-2025 Month RESET

Status	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
Generated	135	1	0	0	0	0	0	0	0	0	0	0
Completed	79	0	0	0	0	0	0	0	0	0	0	0

Search X

Sr. No.	Request No.	Product	Line	Machine No.	Machine Name	Problem	Date-time	Final Action	Loss Time	WO Status	R.S Status	Actions
1	EL-ENG Sealer Block-01-4-2	Final Assy	ENG Sealer Block-01	EEAT-069	Spray Sealer M/C	sealer m/c stop in auto cycle	05-04-2024 T00:21	Cylinder read switch wire damaged portion cut & its dressing done	120	Closed	Under MTD HOSS Approval	
2	EL-MA2-5-20	Mounting	MA2	EETP-009	Board Flipper	hsh	02-05-2024 T09:16	Asdfqwer DFG dfg	60	Closed	Under MTD HOSS Approval	

Rows per page: 5 rows 1-2 of 2 < >

- Every user get approval request in this type of dashboard and on click of action request-sheet is open.

DENSO Integrated Maintenance Operation System Ashish Gupta(2987)

PM BM MTD KPI

Is Action Temporary? ☐ Yes ☒ No

No. Of TM Attended. Sreekanth Sasi Select

Problem Category ☒ Electrical ☐ Mechanical ☐ Both ☐ Pneumatic

Breakdown Category ☒ Wear ☐ Part-Failure ☐ Design ☐ Obsolete ☐ Human ☐ Process ☐ Communication

Skill

PART NO.	PART NAME	MAKER	QUANTITY	Cost	UPDATE
D-M9BVL	REED SWITCH	SMC INDIA	1	1053	Edit Delete

NOTE:
* IN CASE OF MAJOR BREAKDOWN, IT IS NECESSARY TO GET THE SIGNATURE OF "GM-PRD" & "GM-MTD" IN "CHECKED BY" BOX.
** PART QUALITY RELATED TO MAINTENANCE WORK.

CHECKED BY

* GM-MTD	* GM-PRD	SECTION INCHARGE (PRD)	TEAM LEADER (PRD)

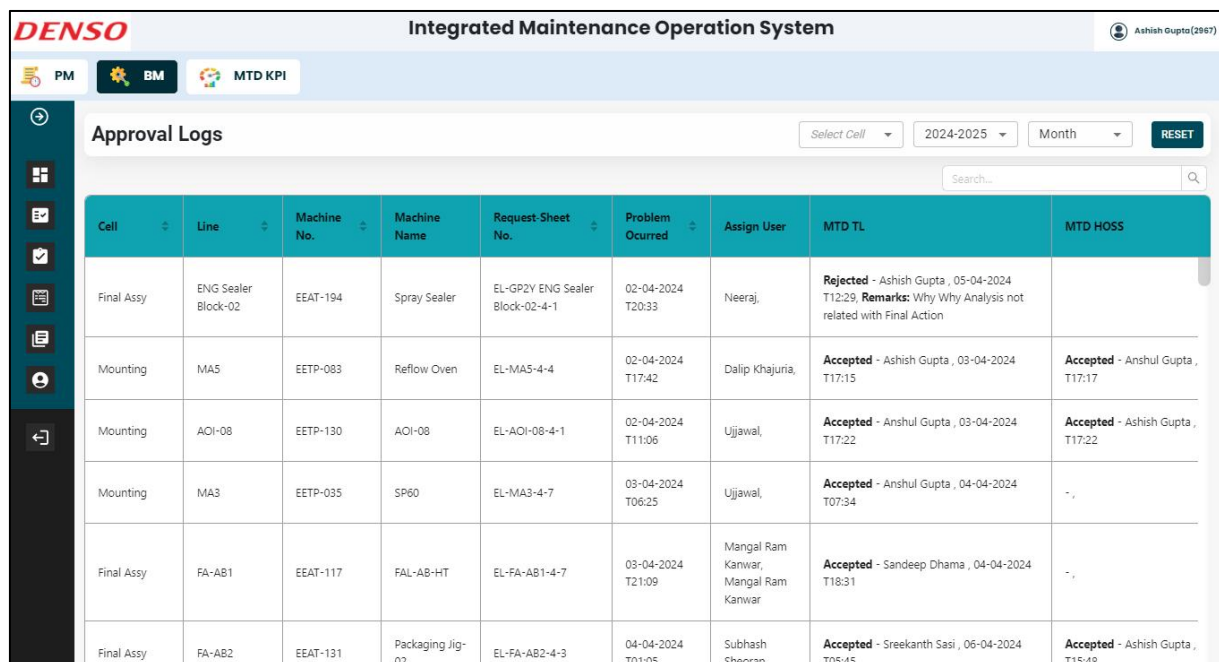
Save Changes

Kindly approve request-sheet.
☐ Yes ☐ No Submit

FO/MTD/01/04/04

- After review whole request-sheet user will able to approve or reject request-sheet. If user is reject the request-sheet then need to add rejected remarks.

3.2.5 Approval logs



Cell	Line	Machine No.	Machine Name	Request Sheet No.	Problem Occurred	Assign User	MTD TL	MTD HOSS
Final Assy	ENG Sealer Block-02	EEAT-194	Spray Sealer	EL-GP2Y ENG Sealer Block-02-4-1	02-04-2024 T20:33	Neeraj	Rejected - Ashish Gupta , 05-04-2024 T12:29, Remarks: Why Why Analysis not related with Final Action	
Mounting	MA5	EETP-083	Reflow Oven	EL-MA5-4-4	02-04-2024 T17:42	Dalip Khajuria	Accepted - Ashish Gupta , 03-04-2024 T17:15	Accepted - Anshul Gupta , T17:17
Mounting	AOI-08	EETP-130	AOI-08	EL-AOI-08-4-1	02-04-2024 T11:06	Ujjawal	Accepted - Anshul Gupta , 03-04-2024 T17:22	Accepted - Ashish Gupta , T17:22
Mounting	MA3	EETP-035	SP60	EL-MA3-4-7	03-04-2024 T06:25	Ujjawal	Accepted - Anshul Gupta , 04-04-2024 T07:34	- ,
Final Assy	FA-AB1	EEAT-117	FAL-AB-HT	EL-FA-AB1-4-7	03-04-2024 T21:09	Mangal Ram Kanwar, Mangal Ram Kanwar	Accepted - Sandeep Dhama , 04-04-2024 T18:31	- ,
Final Assy	FA-AB2	EEAT-131	Packaging Jig-02	EL-FA-AB2-4-3	04-04-2024 T01:05	Subhash Sheoran	Accepted - Sreekanth Sasi , 06-04-2024 T05:45	Accepted - Ashish Gupta , T15:48

- The approval log dashboard exhibits a list of request sheets currently undergoing any approval stage, showcasing their respective statuses, timestamps, and remarks.
- User is easily find out that the request-sheet is in which status and stage.
- Additionally, users can search for request sheets by their unique numbers and filter request sheet counts based on cell/product, line, machine, and year/month.

3.2.6 Other loss data entry (Only MTD member can access)

- MTD users have the capability to input no loss entry data.
- Users are required to select the cell/product first, followed by the line and machine. Based on this selection, MTD users can generate a new entry, which will then be displayed on the master log dashboard.
- Additionally, users can upload files and images as needed.
- Users must select both the "From" and "To" dates and times compulsorily to calculate loss/Breakdown (BD) time accurately.

3.2.7 Master log

DENSO Integrated Maintenance Operation System Ashish Gupta(2887)

PM BM MTD KPI

Master Log Select Cell 2024-2025 Month RESET CSV

Sr. No.	Month	Date	Cell	Line	Machine	Machine No	Shift	Category	Time	Problem	Cause
1	Apr	02-04-2024 T20:33	Final Assy	ENG Sealer Block-02	Spray Sealer	EEAT-194	B	BM	42	Thinner abnormal flow condition	- M - au - Th - cir - ab - Th - cir - Th - pr - frc - De
2	Apr	02-04-2024 T17:42	Mounting	MAS	Reflow Oven	EETP-083	B	BM	28	N2 pressure trouble error coming	- N2 - Sw - N2 - Dr - N2 - N2 - na

- All machine logs including Breakdown Maintenance (BM), Preventive Maintenance (PM), and No Loss entry data alongside each entry..
- Users can filter data based on date, shift, category, whether the action is temporary, who performed the action, and the status of the action.

DENSO Integrated Maintenance Operation System Yogesh Dhyani(131)

PM BM MTD KPI

Master Log Powertrain Gasoline Select Cell 2024-2025 Month RESET CSV

Time	Problem	Cause	Action	Counter Measure	Category	Is Action Temporary?	Done By	Status	View
0	Handle broken	- During Continuous Running - During Continuous Running - During Continuous Running - N/A - N/A	Handle Change with new one	NA	- Problem Category - Mechanical - Breakdown Category - Wear	No	- Yogesh Kumar - Anil Kumar	Completed	
	Master was coming NG during Running	- Air Valve was off - Valve MADE OFF after yesterday production. - NA - NA - NA	Air Valve Open	NA	- Problem Category - Pneumatic - Breakdown Category - Process	No	Yogesh Kumar	Completed	
		MACHINE RUNNING FROM LONG TIME			Problem Category -				

- Users have the capability to filter request sheets according to cell/product, line, and machine. They can opt to view request sheets either by month or year and have the option to download them in CSV format.
- Users can access and review each request sheet and check sheet. Additionally, they have the ability to edit or delete non-loss data entries after reviewing them.

Other Loss BD Entry Form

Close

CELL/PRODUCT, LINE AND MACHINE : Mounting, MA2, Board Flipper, EETP-009

NO-LOSS BD NO:
2024-May-8

MAINT. TYPE
☐ CM
☒ BD with No Loss
☐ Documentation
☐ PRD Support
☐ PED Support

SHIFT
☒ A ☐ B ☐ C

IS ACTION TEMPORARY?
☒ Yes ☐ No

FROM Date & Time:
03-05-2024 03:40 PM

TO Date & Time:
03-05-2024 04:40 PM

TOTAL TIME:
60

Problem Category
☐ Electrical
☒ Mechanical
☐ Both
☐ Pneumatic

Breakdown Category
☐ Wear
☐ Part-Failure
☒ Design
☐ Obsolete
☐ Human
☐ Process
☐ Communication

Skill

MACHINE
STATUS
☒ OK ☐ NG

DONE BY
Ashish Gupta

SUPPORTING TM
Newraj

ATTACHED FILES
Choose Files No file chosen
1714733246807_image.png

PROBLEM	UPDATE	ACTION & COUNTERMEASURE STEPS	STATUS	UPDATE	CAUSE
Problem 1: ASDFG	Edit Delete	Action 1: SDFVGB	☒ OK ☐ NG	Edit Delete	sdfgdxvbsdfg
Problem 2: aswdef	Edit Delete	Action 2: asdfa	☐ OK ☒ NG	Edit Delete	

Add Problem

3.2.8 Admin User

3.2.8.1 Spare order request email (To address) registration

The screenshot displays the DENSO Integrated Maintenance Operation System Admin Dashboard. A modal titled "Email Configuration" is open, allowing an administrator to add or update email addresses for spare order requests. The modal includes fields for "Enter Server-IP", "Enter Port", "From Email Address", and "TO Email Address For Spare Order Request". A "Submit" button is at the bottom of the modal. In the background, the "Admin Dashboard" is visible, featuring a table of email configurations and a sidebar with navigation icons. The "Email Configuration" button in the top right of the dashboard is highlighted with a red box.

Sr. No.	TM No.	TM Name	Email
1	9999	Daily Report	as@gmail.com
2	3501	Toshiyasu Takeuchi	toshiyasu.takeuchi.a2a@ap.denso.com
3	3512	Haruyoshi Kato	haruyoshi.kato.a4b@ap.denso.com
4	1312	Jeetendra Arora	jeetendra.arora.a4y@ap.denso.com
5	37	Anup Shekhar	anup.sherekar.a0x@ap.denso.com
6	1311	Yogesh Dhyani	yogesh.dhyani.a6k@ap.denso.com
7	42	Aman Arya	aman.arya.a6z@ap.denso.com

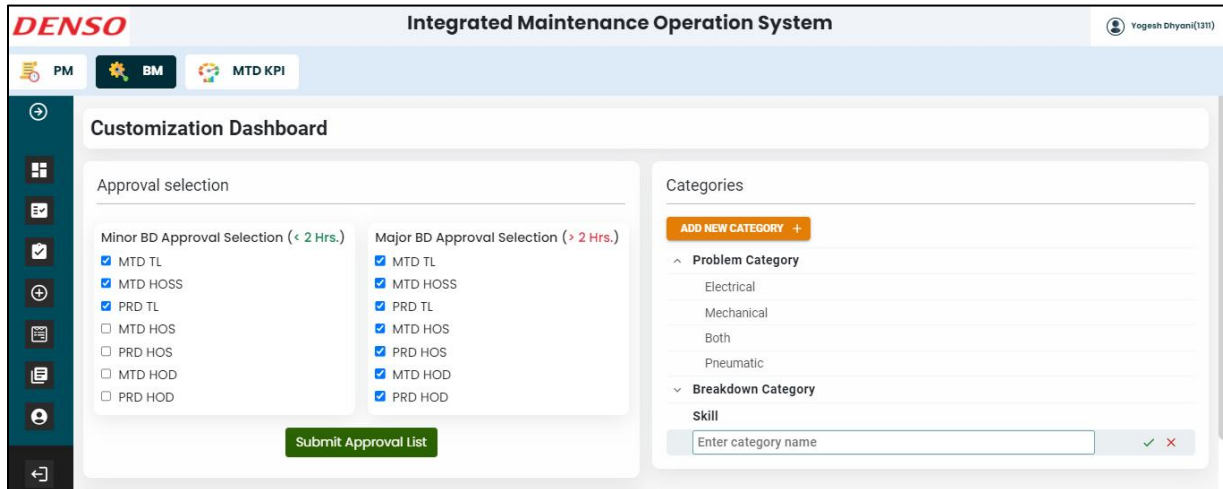
Joining Date	Contact No	Address	Actions
2020-01-01	5	5	
2022-01-01	1	Gurgaon	
2022-01-01	12	Gurgaon	
2013-04-01	8800655992	Delhi	
2021-03-30	9818409469	Gurgaon	
2007-01-01	9999248231	Delhi	
1999-01-01	9810241680	Delhi	

- In the email configuration, the administrator will have the ability to add or update email addresses specifically for spare order requests initiated by either TM or TLs/HOSS.

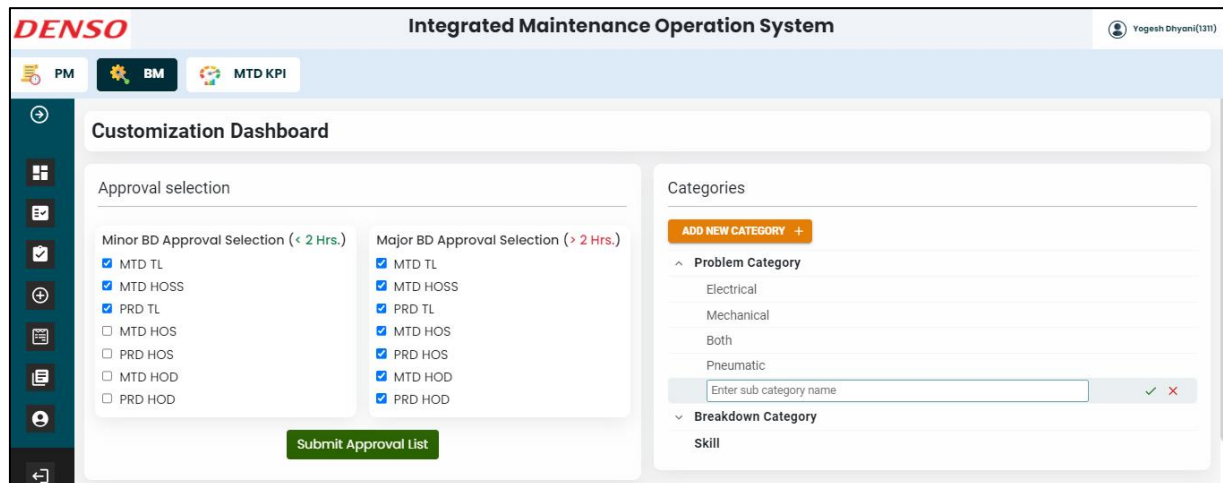
3.2.9 MTD HOD/ MTD HOS

3.2.9.1 Customized dashboard (set dynamic approval, add categories, add shift)

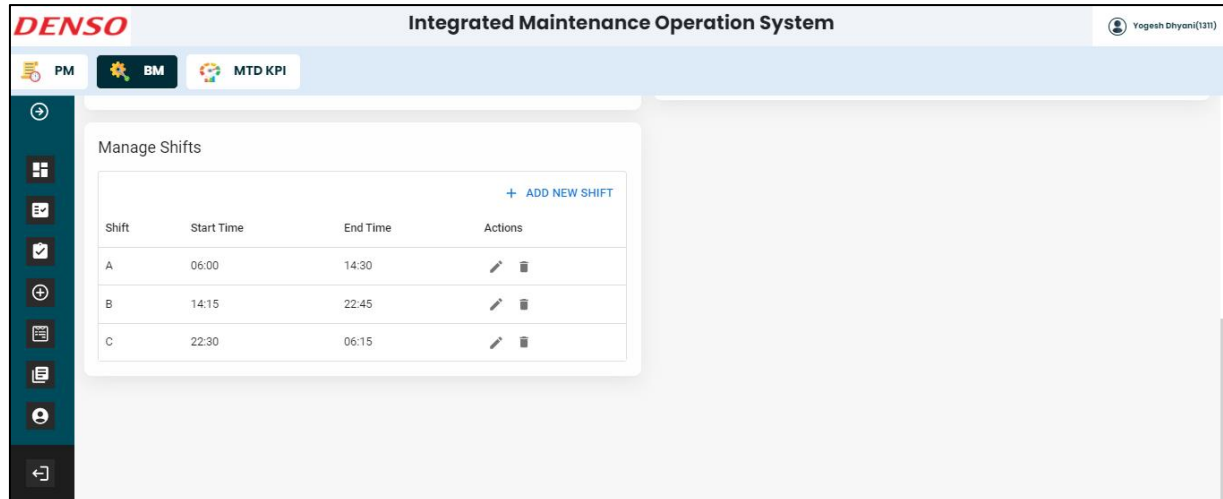
- Before generating a request sheet for a section, MTD HOD needs to establish an approval flow, specify the category, and define the shift for the request sheet.



- HOD users can implement dynamic approval processes for both minor and major cases.
- Note: Initially, MTD TL must be selected as the primary approver. Failure to designate MTD TL as the first approver may result in the approval process not functioning as intended.



- MTD HOD enables the addition of new categories along with their respective sub-categories. Additionally, users have the capability to modify or delete both categories and sub-categories as needed.



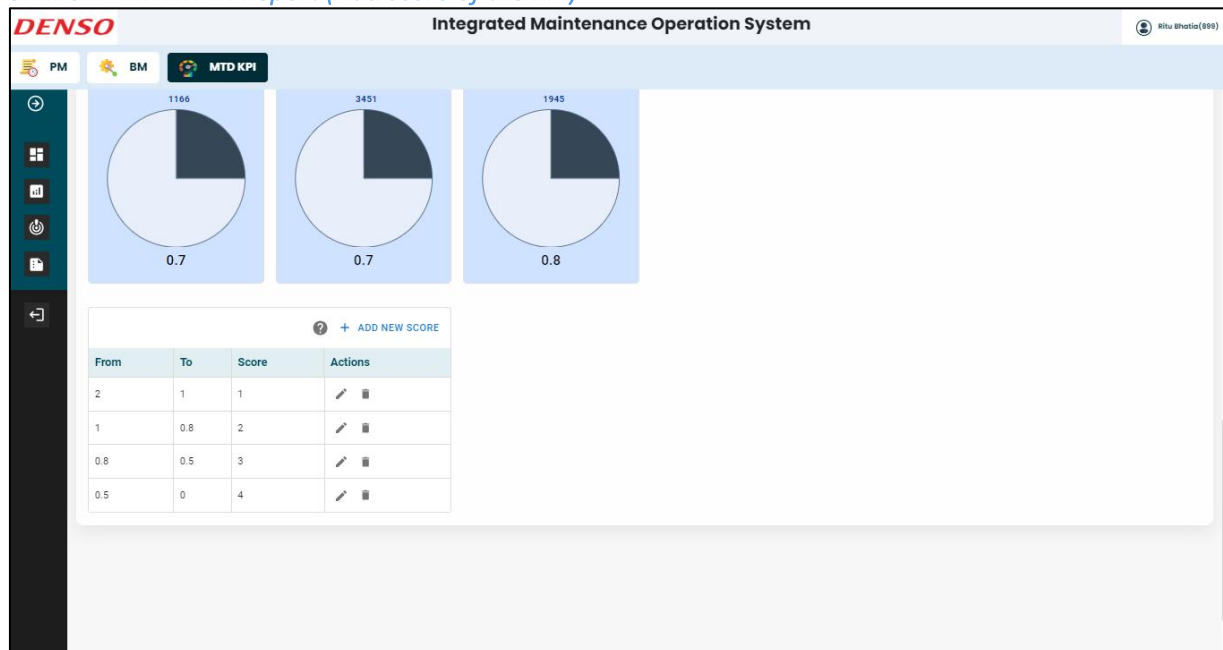
- The MTD HOD is responsible for managing shifts used in request-sheets, including adding, updating, and deleting them as needed.

3.2.10 MTD HOS

3.2.10.1 Approval dashboard

- While request-sheet approval on MTD HOS side, need to add feedback and then approve or reject request-sheet.

3.2.10.2 TM MTTR Report (Add score of the KPI)



- MTD HOS has the capability to add, update, or delete scores within the TM MTTR progress report.

3.2.11 MTD TL/Hoss

3.2.11.1 Request-sheet progress monitoring (Assign request-sheet)

S.No.	Request No.	Product	Line	Machine No.	Machine Name	Problem	Date-time	R.S. Status	Handover To	Final Action	H/O Time Work End	First Time/Repeat	Loss Time	W.O. Status	Actions
1	EL-MA2-S-20	Mounting	MA2	EETP-009	Board Flipper	hgh	02-05-2024 T09:16	Under MTD Hoss Assigned		Asdfower DFG dfg	02-05-2024 T10:16	Repeat	60	Closed	
2	EL-MA2-4-19	Mounting	MA2	EETP-044	SPI	QWERFG	24-04-2024 T16:10	Under MTD Hoss Approved		Gyuir kxvSDFdety fdf wSERFG	24-04-2024 T16:33	First Time	143	Closed	
3	EL-MA2-4-18	Mounting	MA2	EETP-044	SPI	QWERFG	24-04-2024 T16:10	Generated						Open	
4	EL-MA2-4-17	Mounting	MA2	EETP-044	SPI	PCB transport error	24-04-2024 T16:33	Generated						Open	
5	EL-MA2-4-	Mounting	MA2	EETP-044	SPIG-2			Generated						Open	

- The request-sheet generated by the PRD user will be visible on the MTD TL/HOSS dashboard.
- Subsequently, the MTD TL will allocate the TM's to the request-sheet for task assignment. Once the TM's are assigned, the status of the request-sheet will show as "Assigned," and the work order status will default to open.
- Once the request-sheet is assigned to the TM's, it will appear on the assigned TM's dashboard. Other TM's will not be able to view the request-sheets assigned to other TM's.

3.2.11.2 Approval Dashboard

- In this dashboard, MTD TL needs to determine whether the received request-sheet pertains to a minor breakdown (BD) or a major breakdown..
 - If the breakdown is minor, the MTD TL needs to select the appropriate higher authority from the list.
 - If the breakdown is major, the MTD TL needs to verify whether the uploaded sheet is correct and then send approval to the higher authority as determined by the MTD HOD.
- If the MTD TL sends the approval to the higher authority, the status becomes "Under {next department grade} approval".
- If rejected by the MTD TL, the assigned TM will have the ability to make edits and resend for approval.

DENSO Integrated Maintenance Operation System

PM BM MTD KPI

MAINTENANCE REPORT (To be filled by MTD)

REQUEST RECEIVED
MTD S.I
Sandeep Dhama

MTD TL
Sandeep
Dhama

WORK STARTED DATE & TIME: 23-04-2024 09:19 AM

WORK ENDED DATE & TIME: 23-04-2024 09:34 AM

SECTION IN-CHARGE
MTD HOSS
Please select

PROBLEM
Problem: 1gm Waight NG Coming Lower Side 0.85gm (Spec-0.95 t. -1.05)
Add Problem

UPDATE
Edit Delete

WHY WHY ANALYSIS [ROOT CAUSE]

WHY-1
1 gm Waight NG Coming

WHY-2
Gel Not Dispense Properly As per Stock

WHY-3
Gel Leakage From Main Unit

WHY-4
Suspected Main Pump Seal Damage

WHY-5
Life Is Over

QUALITY CONFIRMED (IPP)
Yes No

****PART QUALITY CHECKED**

PRD
Shamsheel

MTD
Sandeep Dhama

DATA SHEET ATTACHED
Yes No

DRAWING ATTACHED
Yes No

BREAKDOWN
15

ANALYSIS
5

SPARE WAITING
0

REPLACEMENT
0

ADJUSTMENT
10

QUALITY CHECK
0

BREAK
0

MAJOR B/D
Yes No

MINOR B/D
Yes No

FIRST TIME
REPEAT

ACTION & COUNTERMEASURE STEPS
Action t: Take Dummy Shot 3-4 Times

STATUS
OK NG

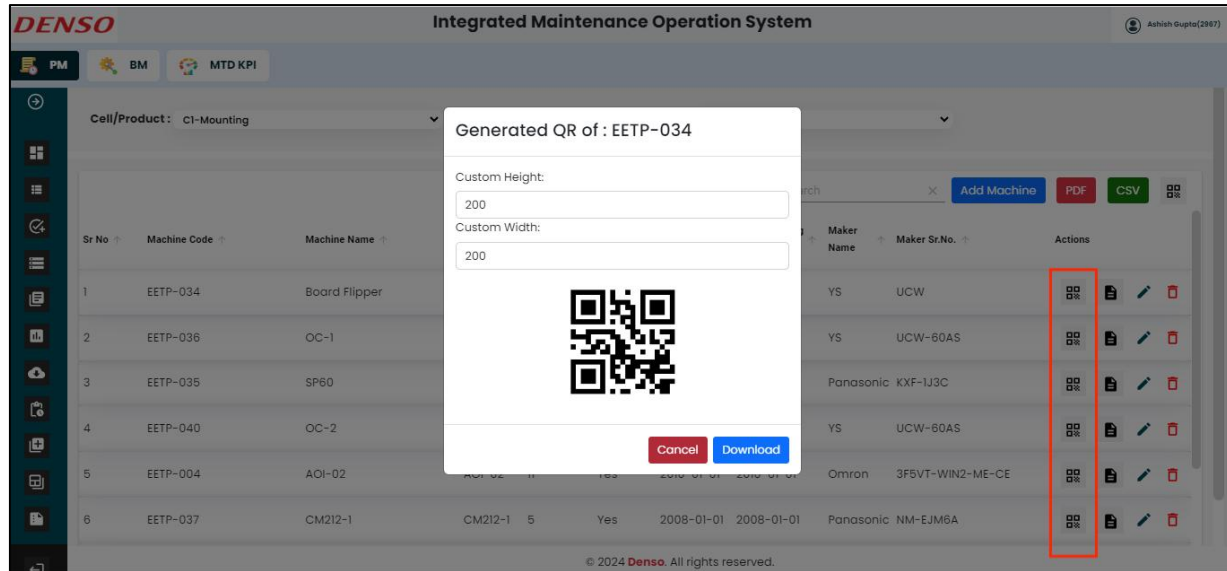
UPDATE
Edit Delete

PREVENTIVE / CORRECTIVE MAINTENANCE
Main Tank Seal Change (T.D-10.08.2024)

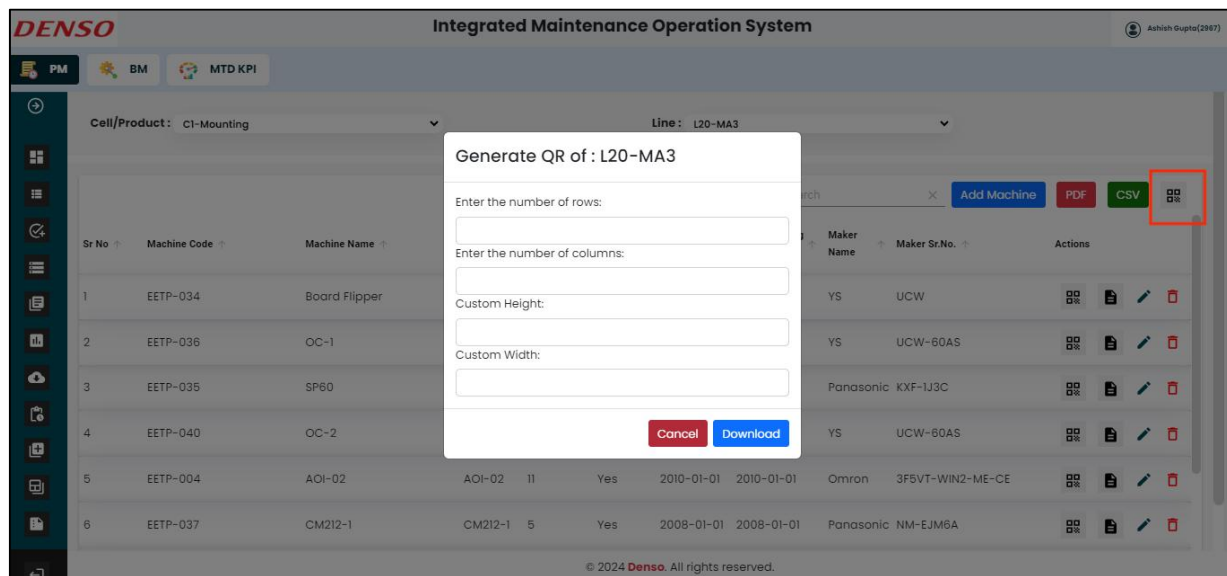
3.2.11.3 QR Code generation (PM Tab)

Sr No.	Machine Code	Machine Name	Machine Nick-Name	Machine Sequence	Proceed for PM?	Installation Date	Manufacturing Date	Maker Name	Maker Sr.No.	Actions
1	EETP-034	Board Flipper	Flipper	2	Yes	2008-01-01	2008-01-01	YS	UCW	[QR Code Icon] [Document Icon] [Edit Icon] [Delete Icon]
2	EETP-036	OC-1	OC-1	4	Yes	2008-01-01	2008-01-01	YS	UCW-60AS	[QR Code Icon] [Document Icon] [Edit Icon] [Delete Icon]
3	EETP-035	SP60	SP60	3	Yes	2008-01-01	0008-01-01	Panasonic	KXF-1J3C	[QR Code Icon] [Document Icon] [Edit Icon] [Delete Icon]
4	EETP-040	OC-2	OC-2	8	Yes	2008-01-01	2008-01-01	YS	UCW-60AS	[QR Code Icon] [Document Icon] [Edit Icon] [Delete Icon]
5	EETP-004	AOI-02	AOI-02	11	Yes	2010-01-01	2010-01-01	Omron	3F5VT-WIN2-ME-CE	[QR Code Icon] [Document Icon] [Edit Icon] [Delete Icon]
6	EETP-037	CM212-1	CM212-1	5	Yes	2008-01-01	2008-01-01	Panasonic	NM-EJM6A	[QR Code Icon] [Document Icon] [Edit Icon] [Delete Icon]
7	EETP-033	YS Loader	Loader	1	Yes	2008-01-01	2008-01-01	YS	PKST-32M3HAS	[QR Code Icon] [Document Icon] [Edit Icon] [Delete Icon]

- In the PM tab, there's a creation dashboard where users can generate customized size QR codes for specific machines.
- Additionally, users can view documents and upload new documents for the machines.



- The image above serves as a reference for generating QR codes for individual machines.



- The image below provides a reference for downloading QR codes for all machines in the line simultaneously.
- Users are required to input the desired number of rows and columns they wish to display, along with the height and width.

3.2.12 PRD TL/HOSS

3.2.12.1 Request-sheet progress monitoring (Generate request-sheet)

Request Sheet Progress Monitoring

Total Requests: 136, Open Requests: 57, Closed Requests: 79

Select Cell, 2024-2025, Month, Req. Sheet Stat..., RESET, GO

Sr. No.	Request No.	Product	Line	Machine No.	Machine Name	Problem	Date-time	R.S Status	Assign	Handover To	Final Action	H/O Time Work End	First Time/Repeat	Loss Time	W.O. Status	Actions
1	EL-MA2-5-20	Mounting	MA2	EETP-009	Board Flipper	hsh	02-05-2024 T09:16	Under MTD HOSS Approval	Jatinder Singh		Asdfqwer DFG dfg	02-05-2024 T10:16	Repeat	60	Closed	
2	EL-MA2-4-19	Mounting	MA2	EETP-044	SPI	QWERFG	24-04-2024 T16:10	Under MTD HOSS Approval	Jatinder Singh		Qyuir xdxSDFDcfv fdfs WSERFG	24-04-2024 T18:33	First Time	143	Closed	
3	EL-MA2-4-18	Mounting	MA2	EETP-044	SPI	QWERFG	24-04-2024 T16:10	Generated							Open	
4	EL-MA2-4-17	Mounting	MA2	EETP-044	SPI	PCB transport error	24-04-2024 T16:01	Generated							Open	
5	EL-MA2-4-16	Mounting	MA2	EETP-094	SPG-2			Generated							Open	

- By clicking the "Generate Request-Sheet" button, PRD users can create new request-sheets for machines.
- Upon clicking the "Generate Request-Sheet" button, a screen is opened where the user can select the machine for which they want to generate a breakdown request-sheet.

Integrated Maintenance Operation System

Back

Electronics

ECU/AB

Mounting

MA2: Loader, Flipper, SPG-2, SPI, OC-1, OC-2, CM212-1, CM212-2, Unloader, CM212-3, Reflow, AOI-1, 3D-1

MA3: Flipper, OC-1, SPG0, OC-2, AOI-02, CM212-1, Loader, CM212-2, 3D-02, CM212-3, Reflow, Unloader

MA4: SPG, Flipper, OC-1, Buffer Conv., OC-2, Reflow, NPM-01, NPM-03, NPM-04, NPM-06, Unloader, Loader, NPM-05, NPM-02

MA5: Loader, Flipper, QR Conv., Buffer Conv., SPI, SPG, NPM-6, Traverser, NPM-7, NPM-8, NPM-9, Traverser-2, Conveyor Cooling, Traverser Pusher, Reflow, Unloader, NPM-5, NPM-4, NPM-1, NPM-3, NPM-2

AOI-5: AOI-05, Scan Conveyor, Loader, 3D-05, Inverter, Unloader

AOI-6: Scan Conveyor, Loader, AOI-06, Inverter, 3D-06, Unloader

AOI-7: PCB Scanner, Loader, AOI-07, PCB Inverter, 3D-07, Unloader

AOI-8: Loader, Scan Conveyor, AOI-08, 3D-08, Inverter, Unloader

AOI-9: Loader, Scan Conveyor, AOI-09, 3D-09, Inverter, Unloader

Laser Line-2: Loader, Unloader, Laser

MA6: LOADER, FLIPPER + CLEANER, QR CONVEYOR, SPG 2, BUFFER CONVEYOR, TRAVERSER-1, DUAL LANE QR CONVEYOR, SPI, TRAVERSER WITH PUSHER, NPM-6, NPM-1, COOLING CONVEYOR, REFLOW, UNLOADER, NPM-2, NPM-3, NPM-4, NPM-5

MA7: LOADER, FLIPPER+CLEANER, SPG2, QR CONVEYOR, SPI, BUFFER CONVEYOR, TRAVERSER-1, NPM-1, NPM-2, DUAL LANE QR CONVEYOR, REFLOW, COOLING CONVEYOR, TRAVERSER WITH PUSHER, UNLOADER

- Based on the machine selection the request-sheet number is auto generated and the PRD user is able to fill-up the request-sheet.

DENSO Integrated Maintenance Operation System

PM BM

Back Machine Details

MAINTENANCE WORK REQUEST/REPORT

PM Status: Current Plan BM: 240.7 Hrs./4 Count CM:

MAINT. TYPE
☐ BM
☐ PM
☐ CM
☐ TPM

PRIORITY CODE
☐ EMERGENCY
☐ IMPORTANT
☐ DATA NEEDED
☐ KAIZEN

REQUEST SHEET (To be filled by PRD)

REQUEST No. EL-MA2-5-21

PROBLEM OCCURRED
 DATE & TIME: 10-05-2024 12:03 PM

SHEET ISSUED
 DATE & TIME: 10-05-2024 12:03 PM

DEPT./LINE
 Mounting/MA2

TL [PRD]
 Harsh Goyal

MACHINE NAME: YS Loader **MACHINE NO.:** EETP-005

PROBLEM FACED: Please select

PRD OBSERVATION FOR THIS PROBLEM BASED ON (5WHY-IHOW)

WHY (5M+1E):

WHERE (Process):

WHEN (Frequency):

WHO (Person):

WHICH (Defect Location):

HOW (Detail/ Observation):

SHIFT
☒ A ☐ B ☐ C

QUALITY RELATED
☐ Yes
☐ No

BREAKDOWN ATTENDED BY

ATTACHED VIDEO OR IMAGES
 Choose Files No file chosen

- It's mandatory to fill in all fields on the request-sheet.
- The shift is automatically selected based on the date and time of the problem occurrence.
- After submit the request-sheet, the request-sheet status is in “Generated” status and this generated request-sheet will be display on all MTD TL/HOSS dashboard for assign the TM to the particular request-sheet.

3.2.13 Operator

3.2.13.1 Request-sheet progress monitoring (Fill request-sheet)

DENSO Integrated Maintenance Operation System

Request Sheet Progress Monitoring

GENERATE REQUEST-SHEET SUMMARY

Total Requests: 1 Open Requests: 1 Closed Requests: 0

Select Cell 2024-2025 Month Assigned RESET GO

St. No.	Request No.	Product	Line	Machine No.	Machine Name	Problem	Date-time	R.S Status	Assign	Handover To	Final Action	H/O Time Work End	First Time/Repeat	Loss Time	W.O. Status	Actions
1	EL-Spot Block-1-4-7	Final Assy	Spot Block-1	EEAT-096	Spot-01	Soldering robot error	16-04-2024 T09:00	Assigned	Jotinder Singh		Final Action	13-05-2024 01:04 PM			Open	✓ X

Rows per page: 1 rows

- Once a request-sheet is assigned to the TM, it appears on their dashboard.
- The TM is responsible for filling in the final action, Hand Over (HO) time, and changing the work order status (Pending, Closed). After changing the WO status, the action button for editing the request-sheet becomes enabled.
- **Note:** The action button remains disabled until all required fields are filled.
- After updating the status of the work order, the request-sheet status is changed to either "Work Order Pending" or "Work Order Closed".

DENSO Integrated Maintenance Operation System

Request Sheet Progress Monitoring

Generate Request-Sheet SUMMARY

Total Requests: 2 Open Requests: 2 Closed Requests: 0

Select Cell 2024-2025 Month Req. Sheet Stat... RESET >= 100 GO Search

Sr. No.	Request No.	Product	Line	Machine No.	Machine Name	Problem	Date-time	R.S Status	Assign	Handover To	Final Action	H/O Time Work End	First Time/Repeat	Loss Time	W.O. Status	Actions
1	EL-MA2-4-19	Mounting	MA2	EETP-044	SPI	QWERFG	24-04-2024 T16:10	Under MTD HOD Approval	Jatinder Singh		Qyuir xdxSDFdcvfdsf WSERFG	24-04-2024 T18:33	First Time	143	Closed	
2	EL-Spot Block-1-4-8	Final Assy	Spot Block-1	EEAT-096	Spot-01	Soldering robot error	16-04-2024 T09:00	Fill Sheet	Jatinder Singh		Sdfgbhnm	16-04-2024 T11:14		134	Closed	

Rows per page: 2 rows 1-2 of 2

- After clicking on the action button, the request-sheet is used to fill in the required data.

DENSO Integrated Maintenance Operation System

Back Machine Details

MAINT. TYPE: ☒ BM ☐ PM ☐ CM ☐ TPM

PRIORITY CODE: ☒ EMERGENCY ☐ IMPORTANT ☐ DATA NEEDED ☐ KAIZEN

REQUEST SHEET (To be filled by PRD)

REQUEST No.: EL-Spot Block-1-4-8

PROBLEM OCCURRED: Soldering robot error

DATE & TIME: 16-04-2024 09:00 AM

SHEET ISSUED: 17-04-2024 T06:39

DEPT./LINE: Final Assy/Spot Block-1

TL [PRD]: Ashish Goyal

MACHINE NAME: Spot-01 MACHINE NO.: EEAT-096

PROBLEM FACED: Soldering robot error Please select

PRO OBSERVATION FOR THIS PROBLEM BASED ON (5WHY-IHOW): Machine stopped after changeover

WHY (5M+IE): No

WHERE (Process): Spot-1 machine

WHEN (Frequency): 100

WHO (Person): Srimanta

WHICH (Defect Location): NA

HOW (Detail/ Observation):

SHIFT: ☒ A ☐ B ☐ C

QUALITY RELATED: ☐ Yes ☒ No

BREAKDOWN ATTENDED BY: Jatinder Singh,,

- The assigned TM is also update the PRD Data if he/she will be needed.

DENSO Integrated Maintenance Operation System Justinder Singh(788)

PM BM MTD KPI

MAINTENANCE REPORT (To be filled by MTD)

REQUEST RECEIVED
MTD S.I
Ashish Gupta

MTD TL
Please select
Srekanth Sasi
Anshul Gupta
Dipshikha
Sandeep Dharma
Ashish Gupta
Masaru Besho

WORK STARTED DATE & TIME: 13-05-2024 02:34 PM WORK ENDED DATE & TIME: 16-04-2024 09:34 AM SECTION IN-CHARGE
MTD HOSS

PROBLEM UPDATE

WHY WHY ANALYSIS [ROOT CAUSE]

WHY-1

WHY-2

WHY-3

WHY-4

WHY-5

QUALITY CONFIRMED (IPP) ☐ Yes ☐ No

**PART QUALITY CHECKED

PRD Please select MTD Please select

DATA SHEET ATTACHED ☐ Yes ☒ No

DRAWING ATTACHED ☐ Yes ☐ No

BREAKDOWN 34 ANALYSIS 0 SPARE WAITING 0 REPLACEMENT 0

ADJUSTMENT 0 QUALITY CHECK 0 BREAK 0

MAJOR B/D ☐ Yes ☒ No ☐ FIRST TIME ☐ REPEAT

MINOR B/D ☒ Yes ☐ No

DENSO Integrated Maintenance Operation System Justinder Singh(788)

PM BM MTD KPI

ACTION & COUNTERMEASURE STEPS STATUS UPDATE

Add Action

PREVENTIVE / CORRECTIVE MAINTENANCE

YOKOTENKAI

Is Action Temporary? ☐ Yes ☐ No

No. Of TM Attended. Select

Problem Category ☐ Electrical ☐ Mechanical ☐ Both ☐ Pneumatic

Breakdown Category ☐ Wear ☐ Part-Failure ☐ Design ☐ Obsolete ☐ Human ☐ Process ☐ Communication

Skill

CHANGED PARTS

PART NO. PART NAME MAKER QUANTITY Cost UPDATE

Add Part

NOTE:
* IN CASE OF MAJOR BREAKDOWN, IT IS NECESSARY TO GET THE SIGNATURE OF 'GM-PRD' & 'GM-MTD' IN 'CHECKED BY' BOX.
** PART QUALITY RELATED TO MAINTENANCE WORK.

CHECKED BY

* GM-MTD * GM-PRD SECTION INCHARGE (PRD) TEAM LEADER (PRD)

Save Changes Send For Approval

FO/MTD/01/04/04

- The assigned TM must complete all mandatory fields, including problems, breakdown times, reasons, attaching data sheets and drawings, specifying categories, and providing details on attendance numbers and parts.
- In the case of a minor breakdown, there's no obligation to upload sheets.

DENSO Integrated Maintenance Operation System

REQUEST RECEIVED
MTD S.I.
Ashish Gupta

MTD TL
Please select

MAINTENANCE REPORT (To be filled by MTD)

WORK STARTED: 06-05-2024 01:14 PM

DATE & TIME: 06-05-2024 01:14 PM

WORK ENDED: 16-04-2024 11:14 AM

DATE & TIME: 16-04-2024 11:14 AM

SECTION IN-CHARGE
MTD HOSS MTD HOS

PROBLEM
Add Problem

UPDATE

WHY WHY ANALYSIS [ROOT CAUSE]

WHY-1

WHY-2

WHY-3

WHY-4

WHY-5

QUALITY CONFIRMED (IPP)
Yes No
This field is required !

**PART QUALITY CHECKED

PRD
Please select
Please select
partQualityCheckedByPRD

MTD
Please select
Please select
partQualityCheckedByMTD

DATA SHEET ATTACHED
Yes No
Choose File No ...sen
This field is required

DRAWING ATTACHED
Yes No

BREAKDOWN
134

ANALYSIS
0

SPARE WAITING
0

REPLACEMENT
0

ADJUSTMENT
0

QUALITY CHECK
0

BREAK
0

Total time is not valid!

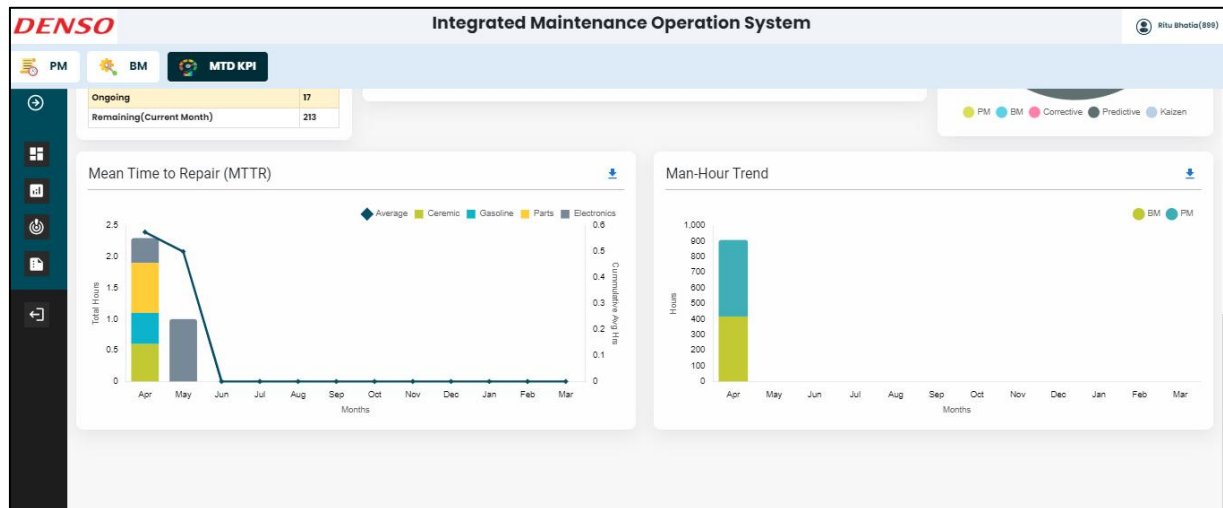
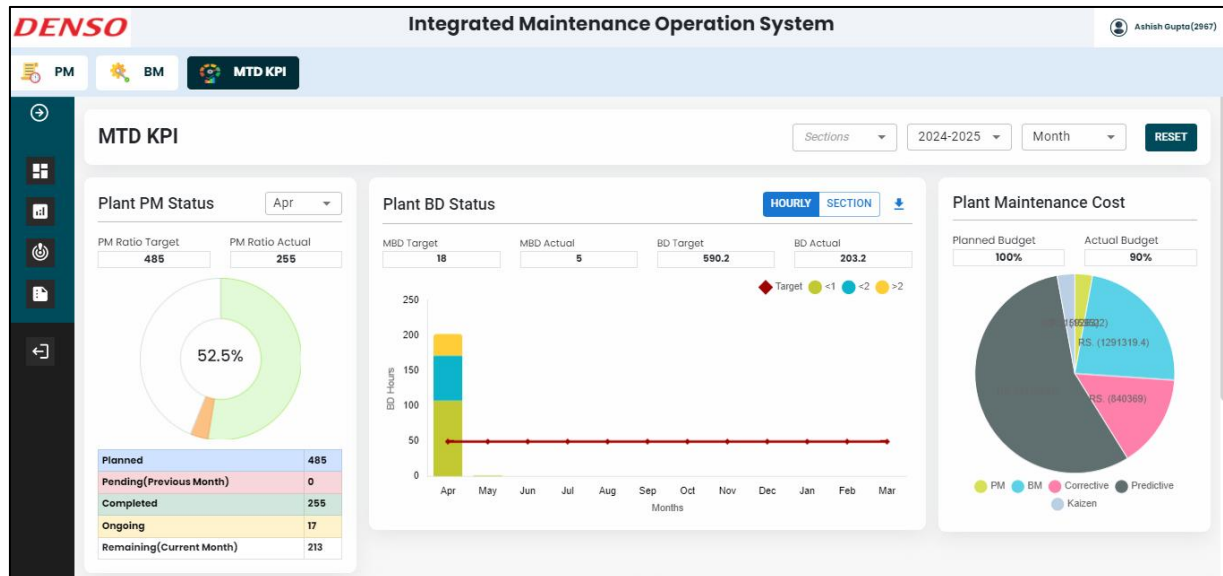
MAJOR B/D
Yes No

MINOR B/D
Yes No

FIRST TIME REPEAT
This field is required !

- In the event of a major breakdown, TM's are required to upload sheets and include preventive and yokotenkai.
- When filling out the MTD section in a request-sheet, TM's need to click on "Save Changes" if they need to save a partially filled request-sheet.
- After completing the entire request-sheet, TM's must select the MTD TL for approval and then click on "Send for Approval".
- The status of the request-sheet after approval is "Under MTD TL approval".
- The request-sheet will be displayed on the approval dashboard of the MTD TL.

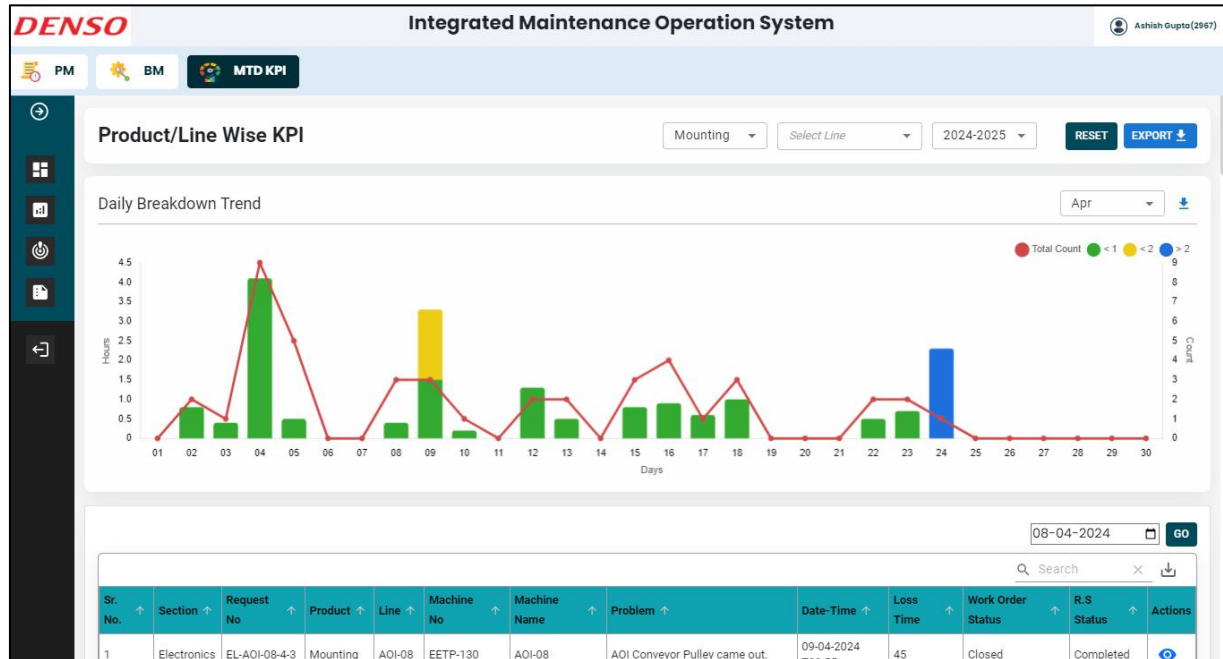
3.2.14 MTD KPI Tab



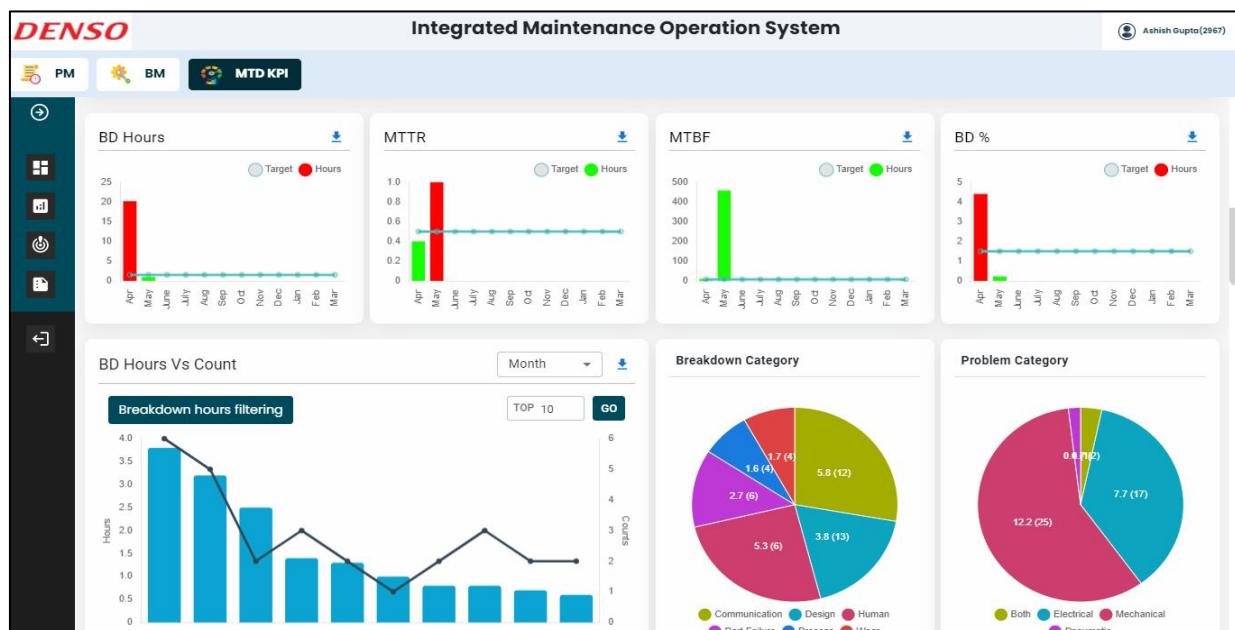
- MTD department users can make decisions based on the following reports.
- The dashboard includes Plant-wise PM Data showcasing planned, pending, completed, ongoing, and remaining tasks. Users can filter the data as needed.
- Additionally, multiple reports such as monthly BD Status and spare consumption are available, displaying data by default on a yearly basis. Plant-wise MTTR across all sections and Man-hour reports for PM and BM are also accessible.

3.2.15 Reports

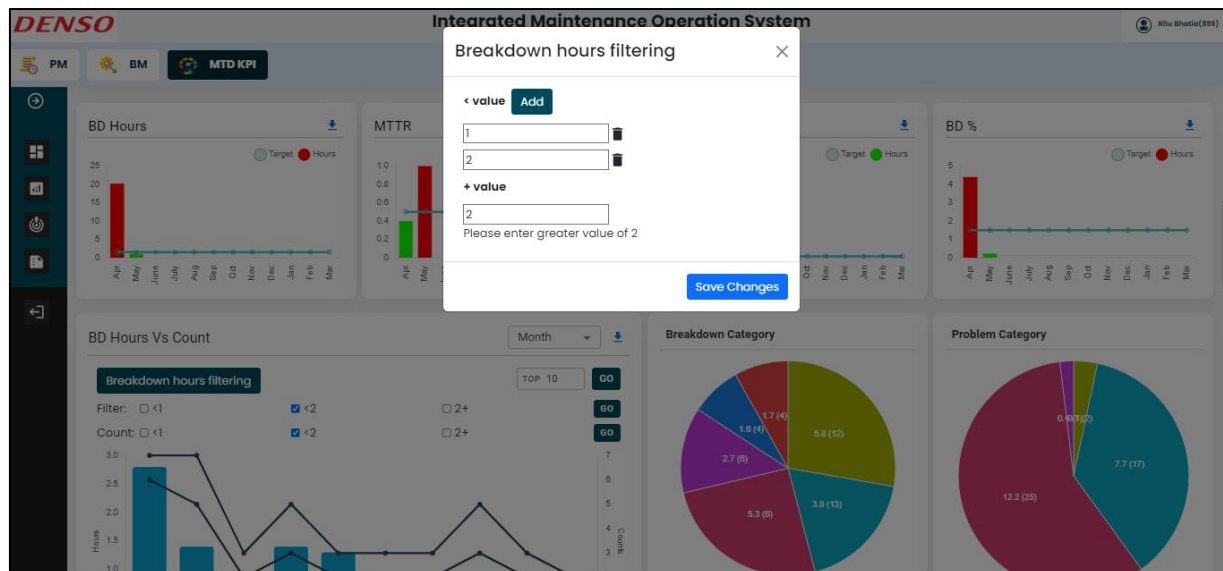
3.2.15.1 Production line wise



- Users can base their decisions on daily breakdown hours and counts, with the option to filter data by months and years.
- Initially, the data is displayed based on cell/product. Users can switch to viewing data based on lines by applying a filter.



- By default, the four types—BD Hours, MTTR, MTBF, BD%—are displayed based on cell/products.
- In this chart, for BD hours, MTTR, and BD%, the bar color is red when the data exceeds the target, and green when it's below the target. Conversely, for MTBF, the color is green when the data is above the target and red when it's below.
- **Note:** The data in this chart is inaccurate if the target is not set for each line and cell/product.
- In the BD hours vs count chart, each machine's data is initially displayed for the top ten machines. If users wish to view more than ten machines, they can enter a value in the text field and click on the "Go" button.
- The pie chart represents the data categorized based on the information entered in every generated request-sheet.



- If users wish to apply multiple filters based on their requirements, such as viewing data less than a certain value in hours (1, 1.5, 2, etc.), they can add these values into the "< value" text field.
- Adding a "+ value" means filtering out data where the last value of the filter is greater than the entered value. This filters out data greater than the value entered in the "+value" field across all machine data.

DENSO Integrated Maintenance Operation System

PM BM MTD KPI

From Date: dd-mm-yyyy To Date: dd-mm-yyyy GO RESET

Search...

Section	Request No	Product	Line	Machine No	Machine Name	Problem
Electronics	EL-MA2-5-20	Mounting	MA2	EETP-009	Board Flipper	
Electronics	EL-MA2-4-19	Mounting	MA2	EETP-044	SPI	
Electronics	EL-MA5-4-7	Mounting	MA5	EETP-073	YS Loader	
Electronics	EL-MA6-4-8	Mounting	MA6	EETP-115	BOARD FLIPPER + CLEANER	
Electronics	EL-MA4-4-26	Mounting	MA4	EETP-053	SPG	
Electronics	EL-AOI-08-4-4	Mounting	AOI-08	EETP-133	MULTIPLE MAGAZINE UNLOADER	
Electronics	EL-AOI-6-4-14	Mounting	AOI-6	EETP-089	AOI-06	
Electronics	EL-MA4-4-24	Mounting	MA4	EETP-055	Buffer Conveyor	conveyor belt broken
Electronics	EL-MA5-4-6	Mounting	MA5	EETP-086	Multi-Magazine Unloader	card stuck error
Electronics	EL-AOI-6-4-13	Mounting	AOI-6	EETP-090	3D-06	communication error

Search in filters

Select all items

☐ During Server Access Error

☐ chuck reor

☐ During Server Access Error

☐ DURING SERVER ACCESS ERROR

☐ UNLOADER STOCKER NOT DOWN

☐ pcb carry in reor

☐ Head Overload Anomaly Error

☐ Camera Height Error

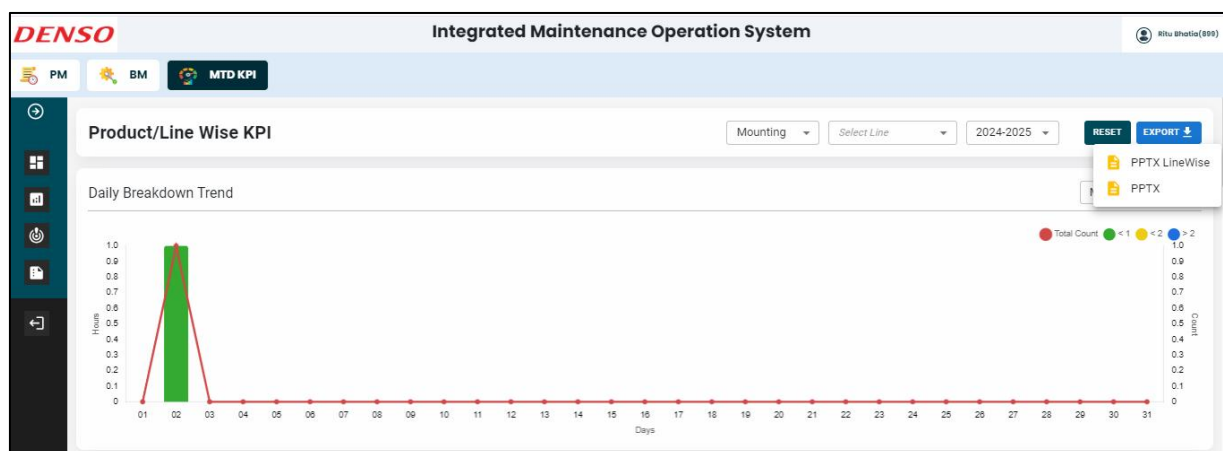
☐ AOI Conveyor belt removed.

☐ N2 pressure trouble error coming

Reset

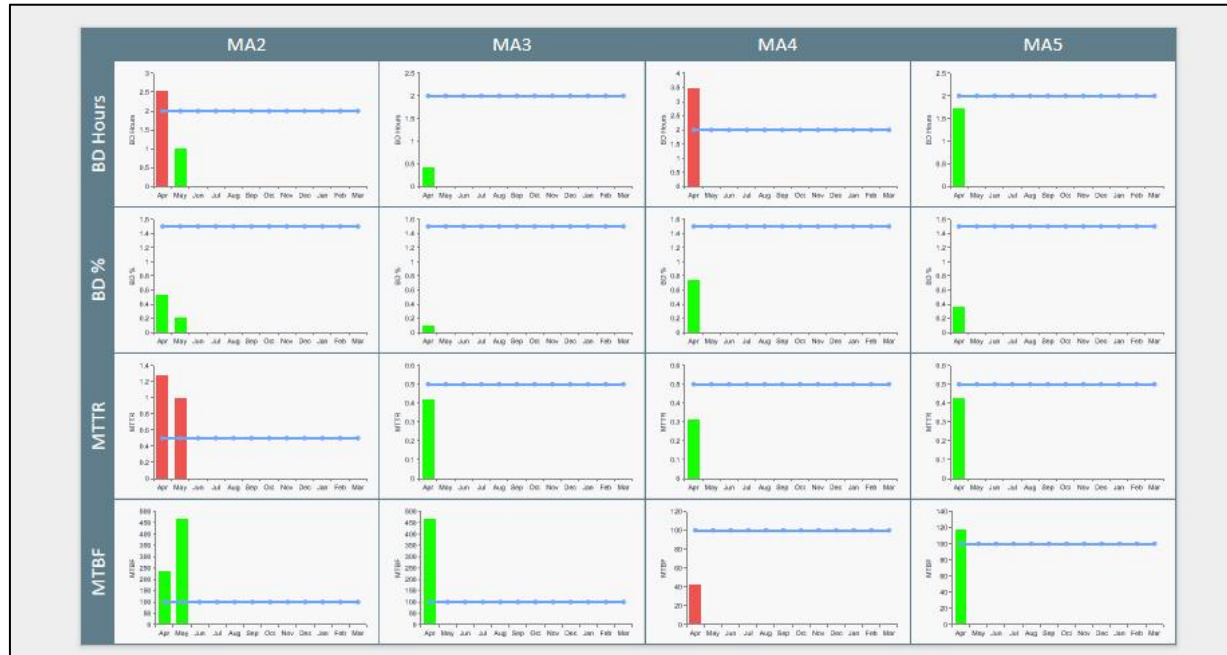
OK

- The table displays all request-sheets generated in the current year, and users can filter out data based on the nature of the problem.
- Users can select multiple problems, choose all, or opt for any single problem based on their preference.
- Users can filter out data by selecting a date range using the "From" and "To" date selection, as well as searching for specific request-sheets.

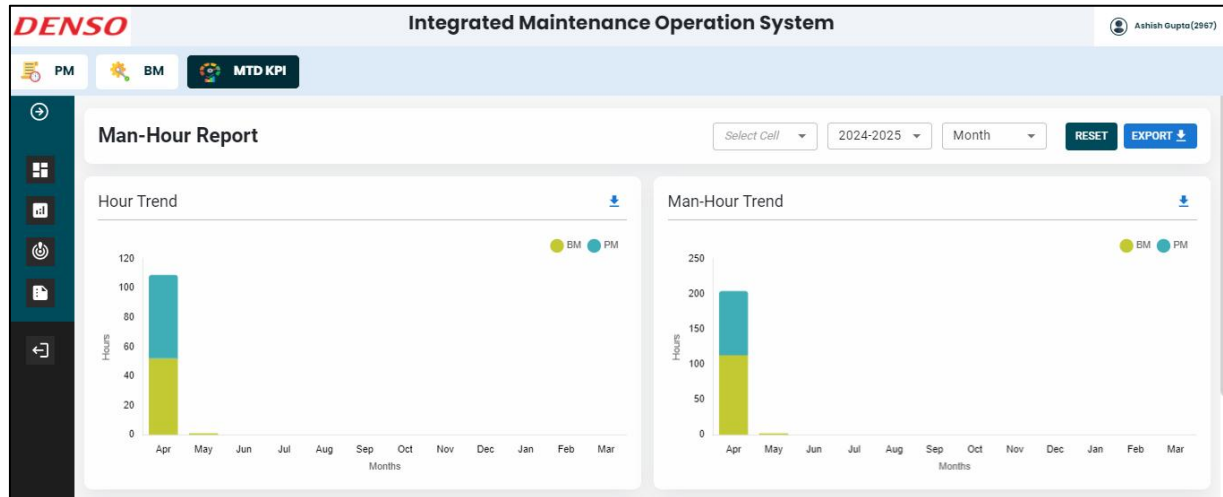


- After click on the export there are two option.
 - The "PPTX Linewise" option is utilized to download a report containing four line-wise charts based on the selection of cell/product.

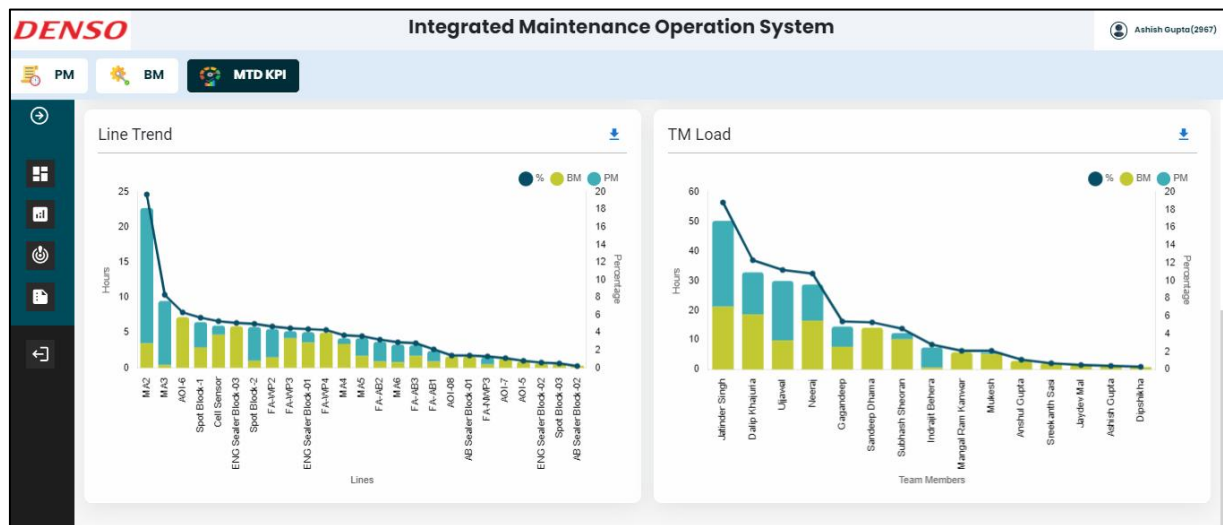
- This presentation contains all lines that belong to the selected cell/product, organized in their respective sequences..



3.2.15.2 Man hour

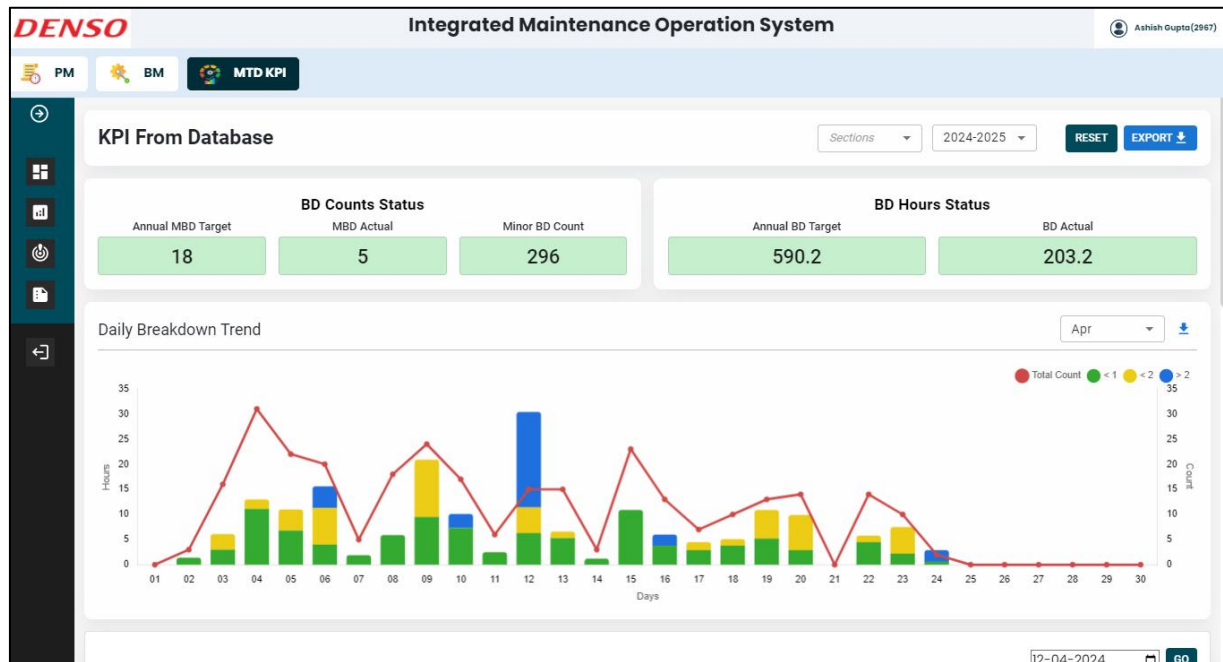


- The hour trend report includes basic hours entered by TM's, along with preventive maintenance (PM) and breakdown maintenance (BM) data..
- In the Man-hour trend report, it includes supporting TM's hours on a month-wise basis.

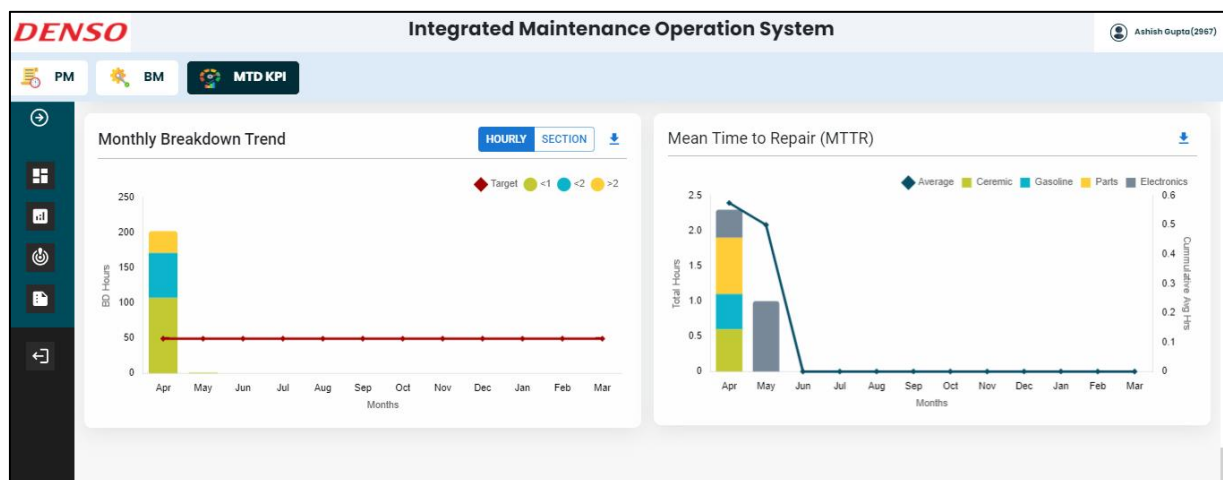


- In the line trend data, only the hours entered by the user are included, encompassing both preventive maintenance (PM) and breakdown maintenance (BM) activities. Additionally, it provides an overall breakdown of line-wise data in percentage.
- In the TM load chart, user-wise hours are displayed for preventive maintenance (PM), breakdown maintenance (BM), and overall time, represented as percentages.
- Users can also filter out data based on cell/products, line, and year/months for further customization and analysis.

3.2.15.3 Daily break-down

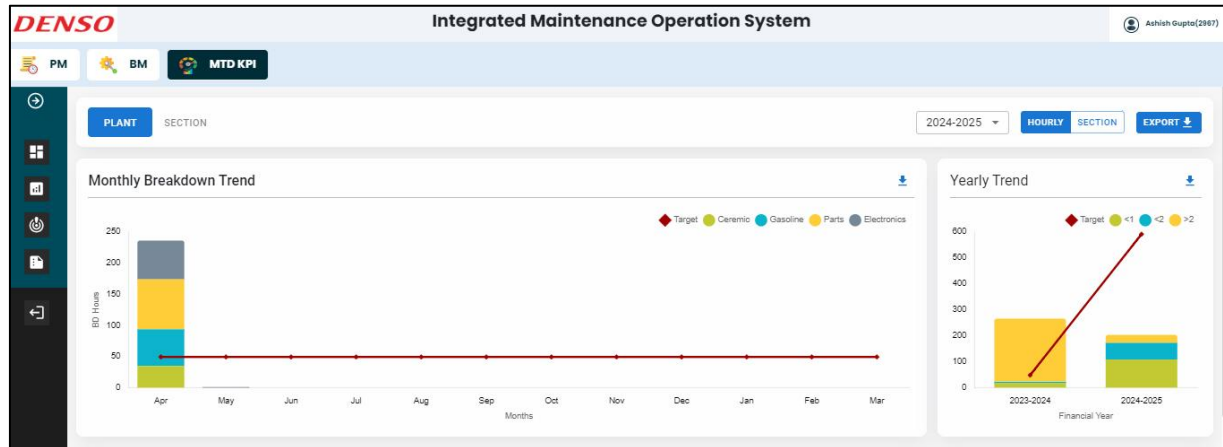


- In this dashboard, targets are displayed on a plant-wide basis, while the actual count is shown against these targets.
- The daily breakdown chart displays each day of the month categorized as less than one, less than two, and greater than two, along with the total counts of breakdowns.

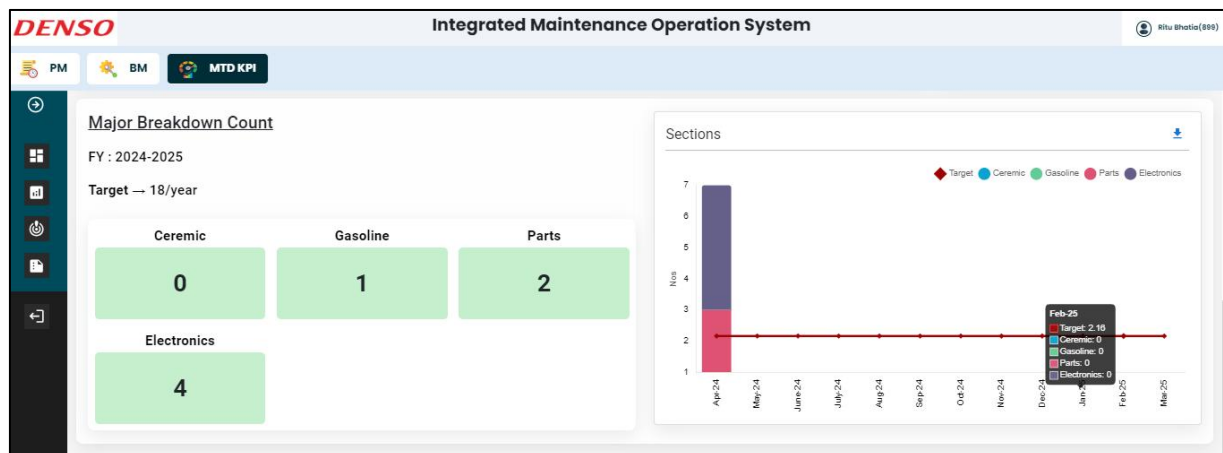


- When selecting dates, users need to click the "Go" button each time to apply their selection..
- The monthly breakdown trend data includes breakdown hours categorized by hours and section, providing a breakdown of data at a lower level of hierarchy.

3.2.15.4 Monthly break-down

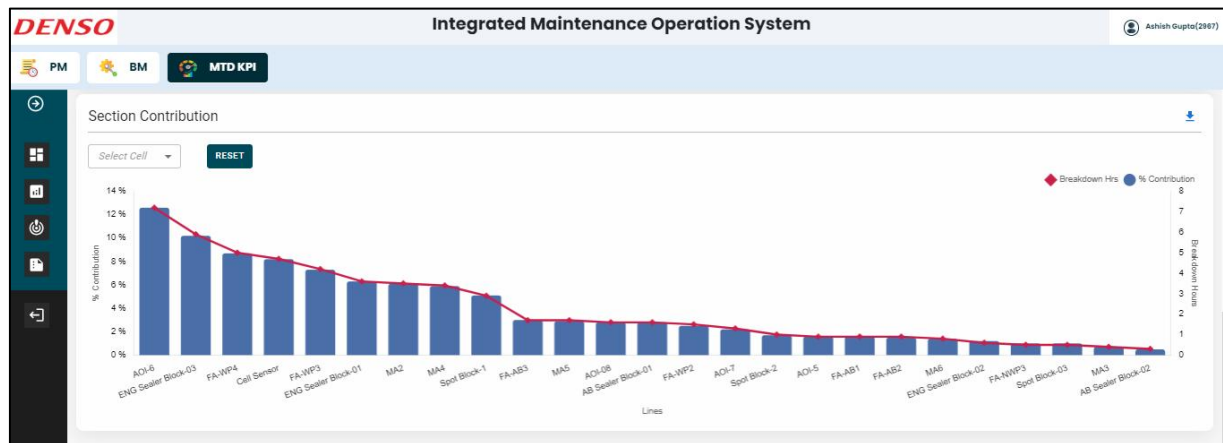
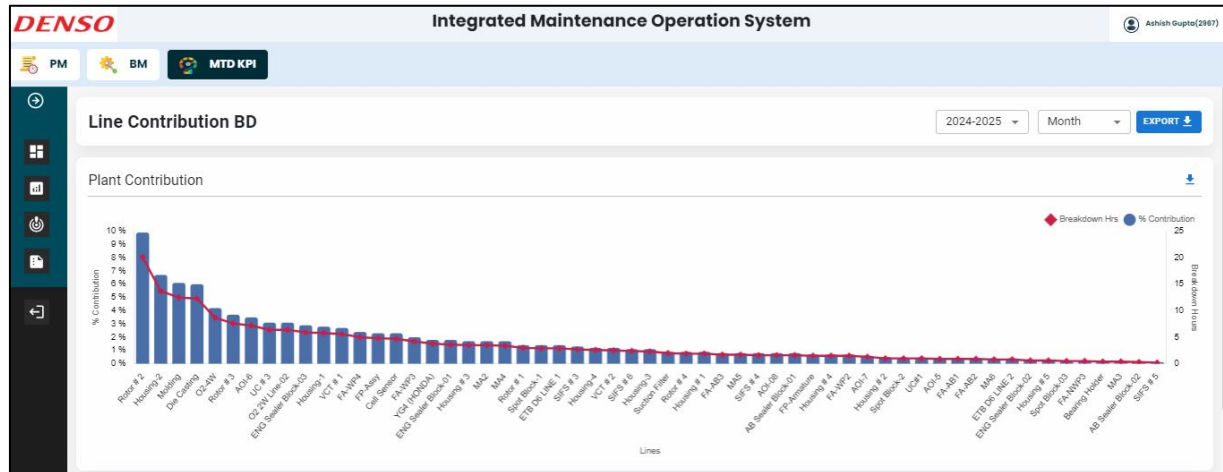


- In this dashboard, by default, the data is displayed at the plant level on an hourly basis. Users can toggle a button to view the data section-wise if desired.
- If users wish to display data by default at the section level, they can click on the section button located near the plant button at the top left. Additionally, users have the ability to filter out data based on cell/product categories.



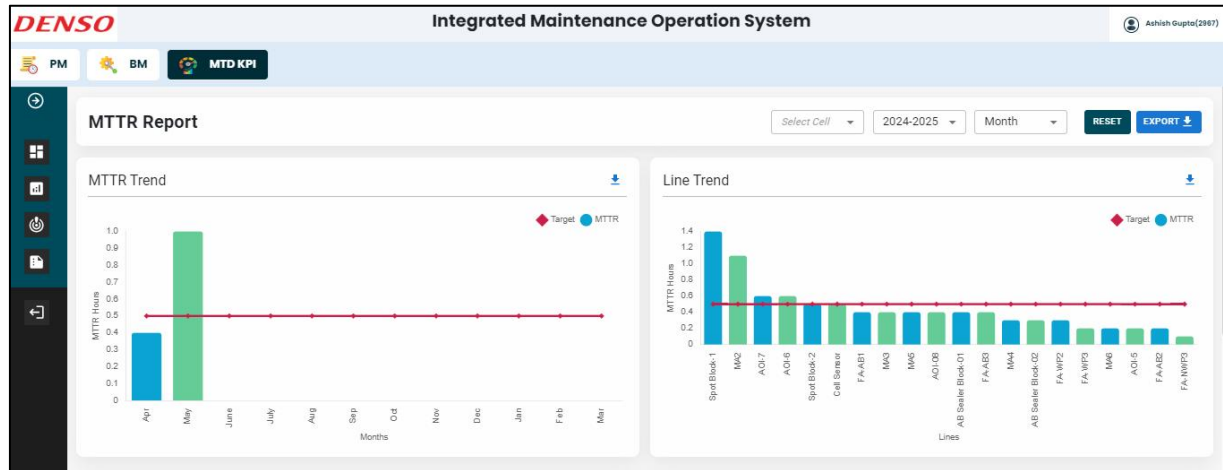
- By default, the major breakdown counts are displayed for all sections within the plants.
- Additionally, it filters out the major breakdown counts based on section and cell-wise counts.

3.2.15.5 Line contribution in percentage

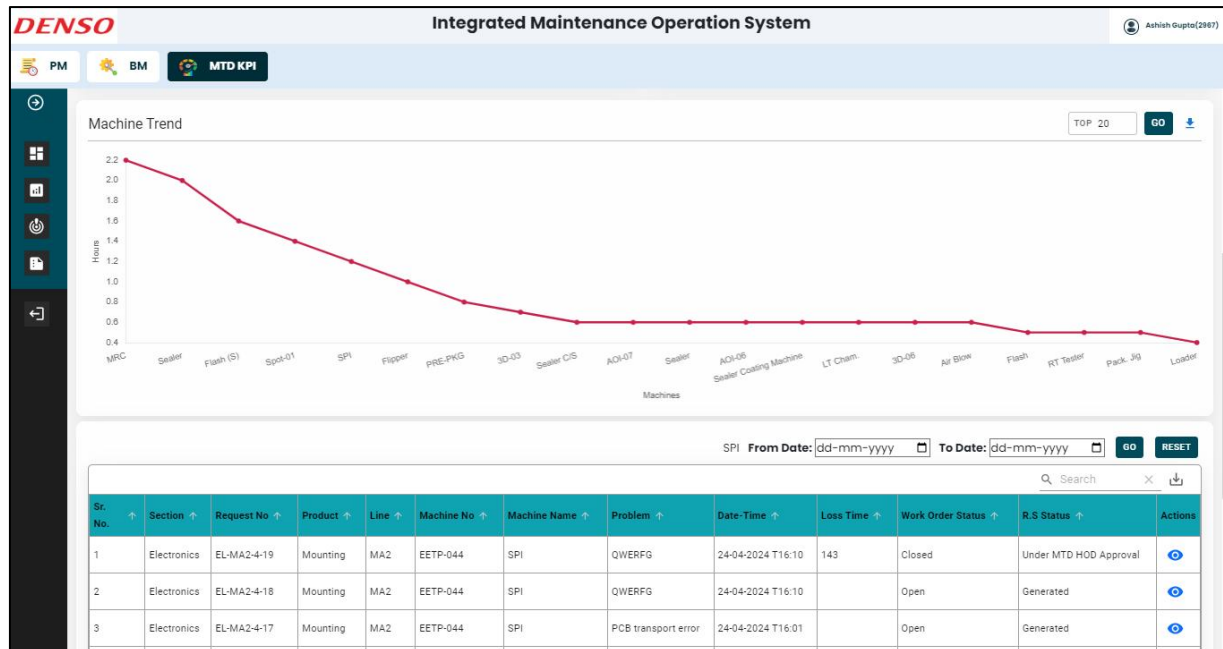


- This dashboard showcases breakdown hours for each line, along with their contribution as a percentage at both plant and section levels.
- Additionally, users can filter out data based on cell/products and year/month for further data display as require..

3.2.15.6 MTTR



- Data is presented section-wise, and users have the option to filter out data based on cell/product and line categories..
- To ensure accurate data, users must first set Mean Time To Repair (MTTR) targets.
- Formula for MTTR Trend (Month wise) :
 - $MTTR = \text{Total BD Hour} / \text{Count}$
 - Target : average of every month
- Formula for Line Trend :
 - Total BD Hour/ Count
 - Target : average of every month up-to current month(In year)
 - ◆ If selected month that the target is count using selected month.



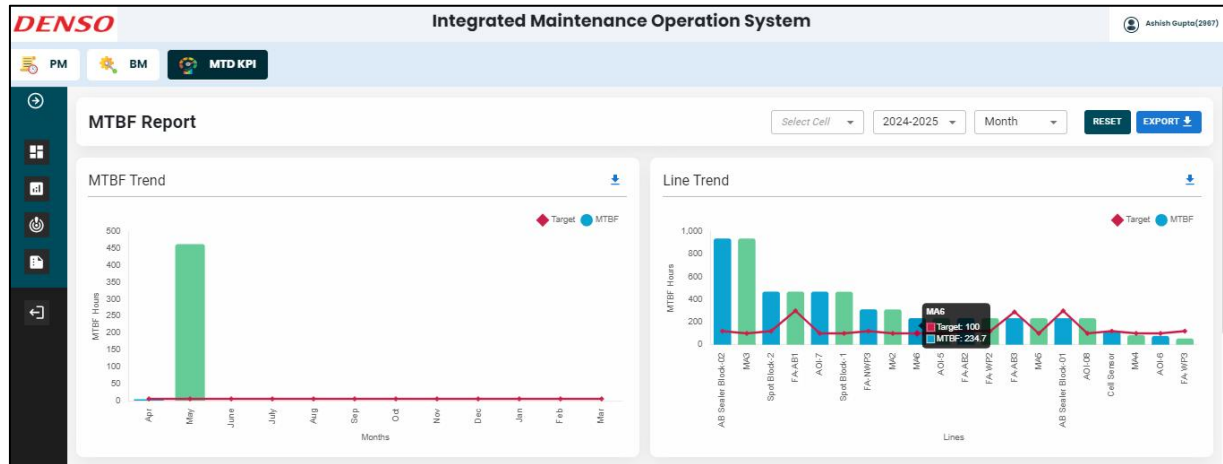
● Formula for Machine Trend :

■ Total BD Hour/ Count

■ Target : average of every month up-to current month(In year)

◆ If selected month that the target is count using selected month.

3.2.15.7 MTBF



- Data is presented section-wise, and users have the option to filter out data based on cell/product and line categories.

- To ensure accurate data, users must first set Mean Time Between Failures (MTBF) targets.

- Formula for MTBF Trend (Month wise) :

- Line :

- ◆ $MTBF = (\text{Production hours} - \text{BD Hours}) / \text{count}$

- ◆ Target : every month

- Cell/product, Section

- ◆ $MTBF = (\text{Average (Production hours)} - \text{sum of (BD Hours)}) / \text{sum of (count)}$

- Average production hrs = Average of all line for particular month production hour target.

- ◆ Target : Average of (MTBF target) all line / count

- Formula for Line Trend :

- $MTBF = (\text{Sum (Production hours)} - \text{sum of (BD Hours)}) / \text{sum of (count)}$

- ◆ Sum (Production hours), sum of (BD Hours)), sum of(count) all are in up-to current month.

- Target : average of every month up-to current month(In year)

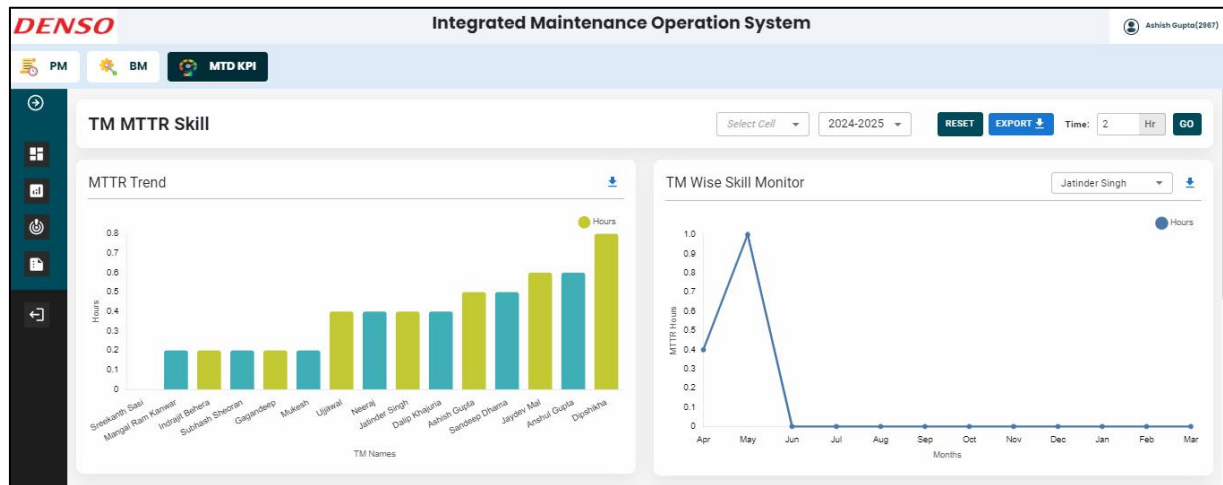
- ◆ If selected month that the target is count using selected month.



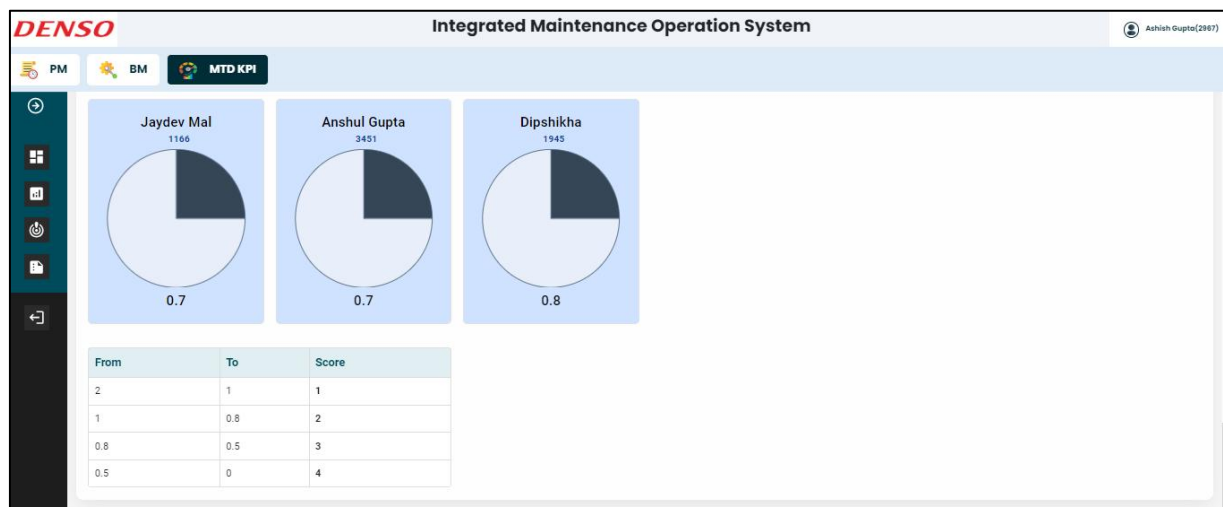
● Formula for Machine Trend :

- $MTBF = (\text{Sum (Production hours)} - \text{sum of (BD Hours)}) / \text{sum of (count)}$
 - ◆ Sum (Production hours), sum of (BD Hours), sum of (count) all are in up-to current month.
- Target : average of every month up-to current month(In year)
 - ◆ If selected month that the target is count using selected month.

3.2.15.8 TM MTTR Skill

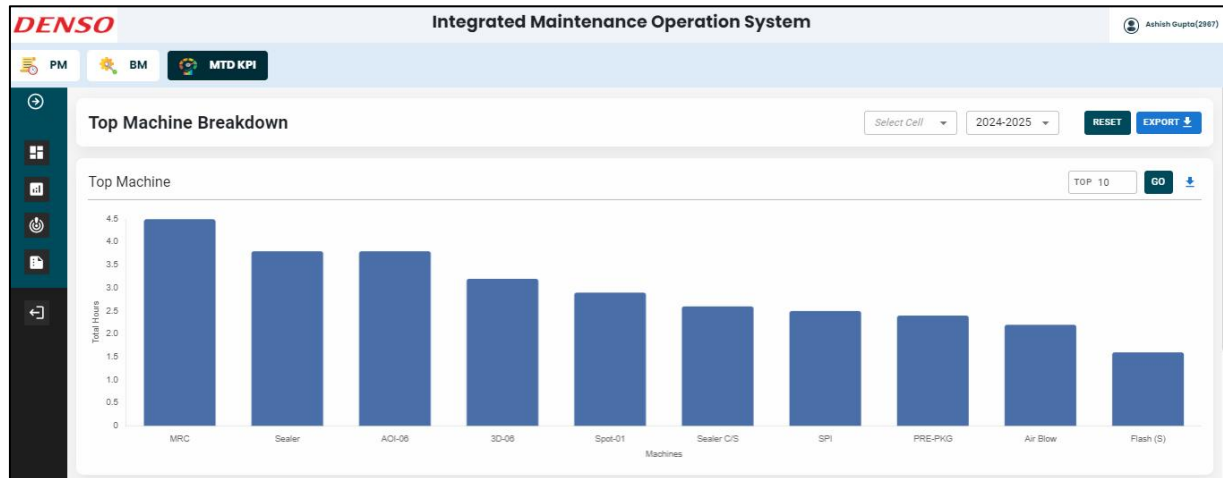


- Trend data for Top Machine Mean Time To Repair (MTTR) displays the performance of every user involved in the breakdown maintenance (BM) process, including support, assignment, and handover responsibilities for request-sheets.
- The TM-wise skill monitor displays MTTR hours versus months, allowing users to filter out other users from the list for focused analysis.
- Additionally, users can apply filters to display only the data they require.
- By default, the display includes instances with less than two hours. Users can modify this threshold by entering a different value into the text box.

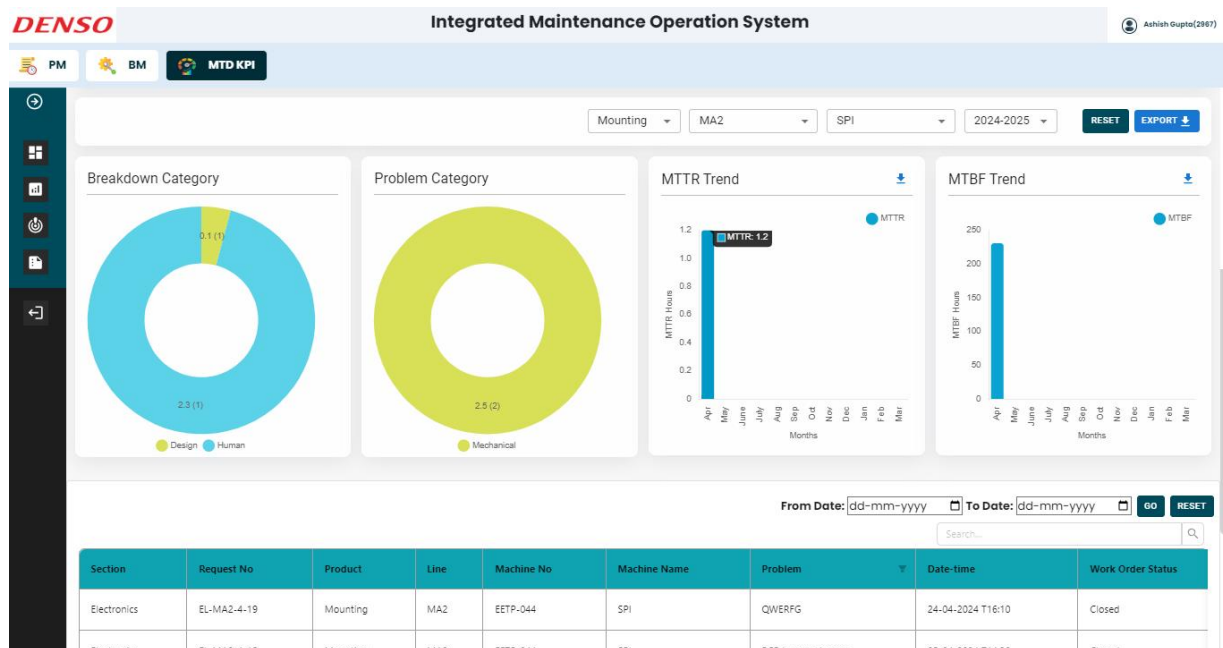


- Users can monitor their Key Performance Indicators (KPIs) in the form of a circle based on the criteria provided by MTD HOS in the table.

3.2.15.9 Top machine break-down



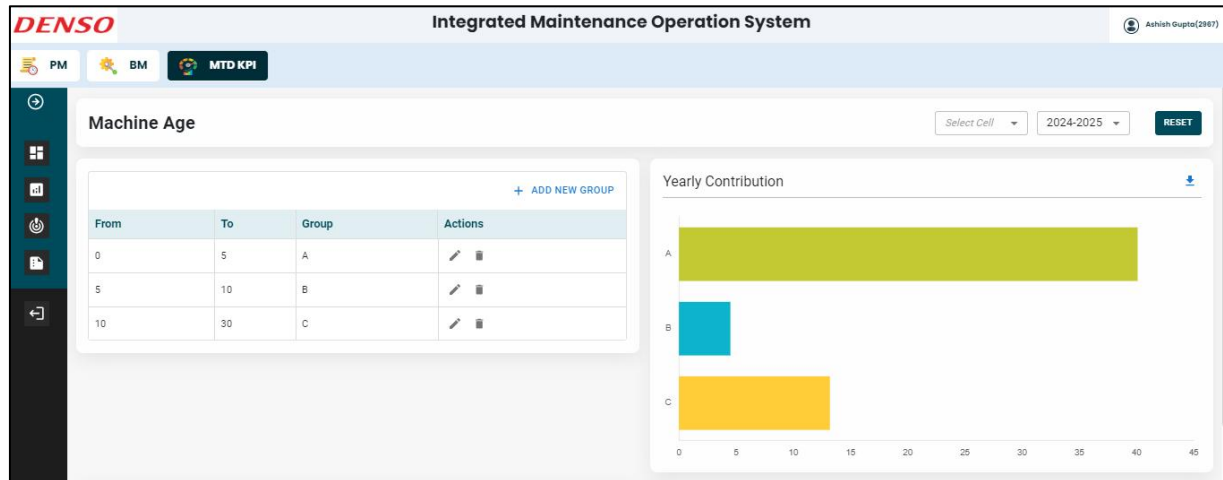
- The top machine chart displays the total hours of each machine.
- By default, this chart displays the top 10 machines with the highest BD hours in descending order. Additionally, users have the capability to adjust the number of machines displayed by changing the value of the top machines setting.



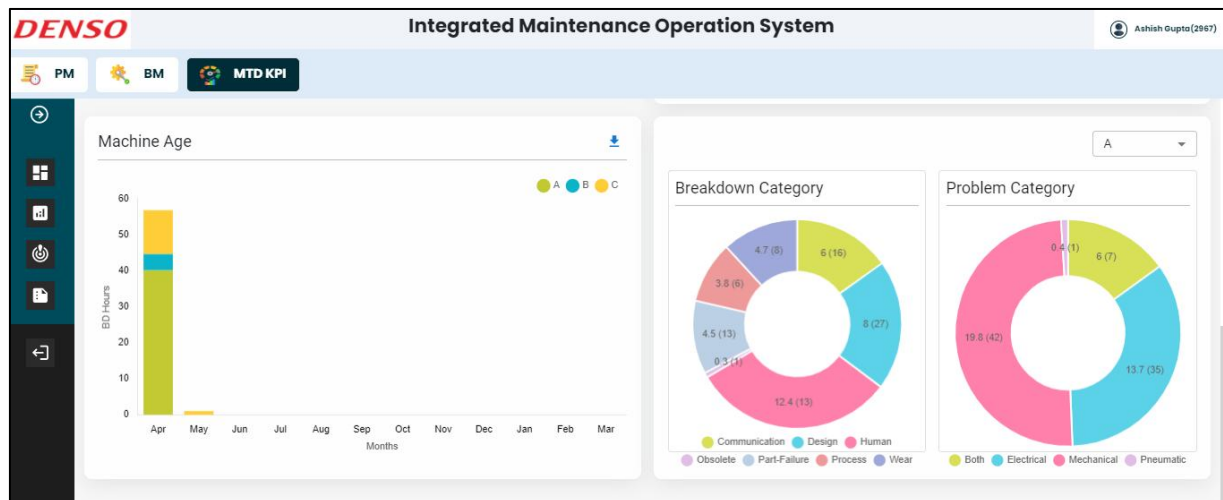
- Here, multiple charts are displayed initially, showcasing data for the first cell, line, and machine by default. Additionally, categorical data as well as trends for MTTR and MTBF are presented.

- If users wish to view data for other machines, they can change the filter value. Upon selecting a machine from the filter, all relevant request-sheets related to the chosen machine are displayed in the table below.

3.2.15.10 Machine Age



- Additionally, by default, the data is categorized, and the initial display showcases the first category from the list.
- The yearly contribution chart display group vs BD hours.



- Machine age wise data display data as group vs months.
- Additionally, data is categorized by default, with the first category from the list being displayed initially.

3.2.16 Target dashboard (MTD HOD/ HOS/ TL-HOSS)

DENSO Integrated Maintenance Operation System

PM BM MTD KPI

Set Target (MBD Count, PRD Hrs., BD Hrs.)

Mounting Select Line 2024-2025 RESET

	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
Major BD Hours	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75

SUBMIT TARGET

- Users have the capability to add monthly Major Breakdown (MBD) targets for individual cells/products. These targets are utilized in the Monthly Breakdown Trend Report for analysis.

DENSO Integrated Maintenance Operation System

PM BM MTD KPI

Set Target (MBD Count, PRD Hrs., BD Hrs.)

Mounting MA2 2024-2025 RESET

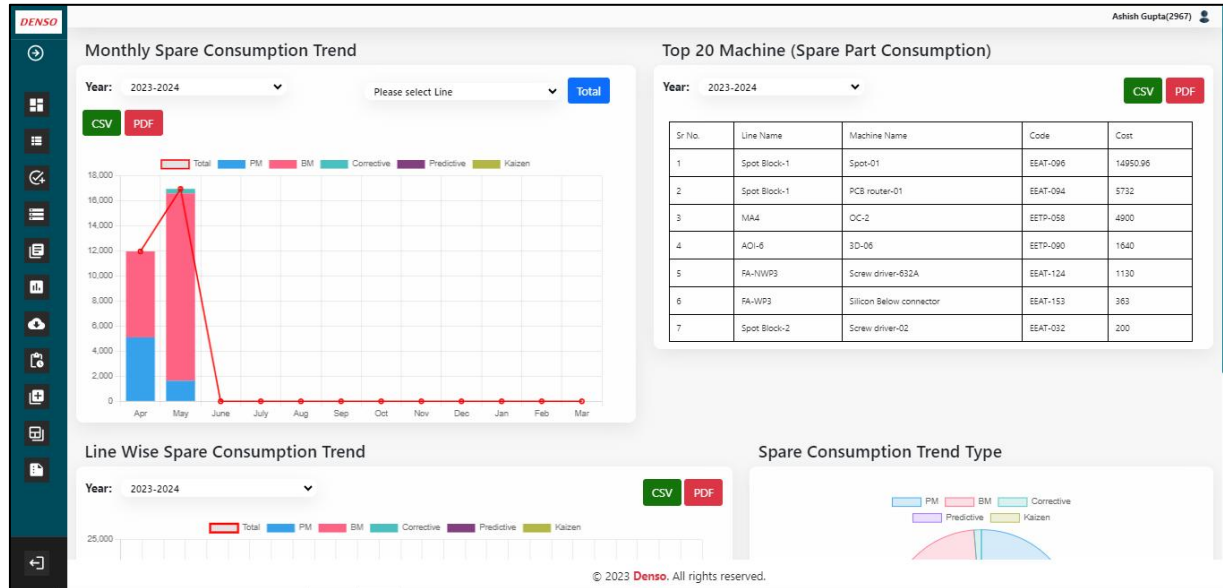
	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
Production Hours	470	470	470	470	470	470	470	470	470	470	470	470
BD Hours	2	2	2	2	2	2	2	2	2	2	2	2
MTTR	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
MTBF	100	100	100	100	100	100	100	100	100	100	100	100
BD %	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5

SUBMIT TARGET

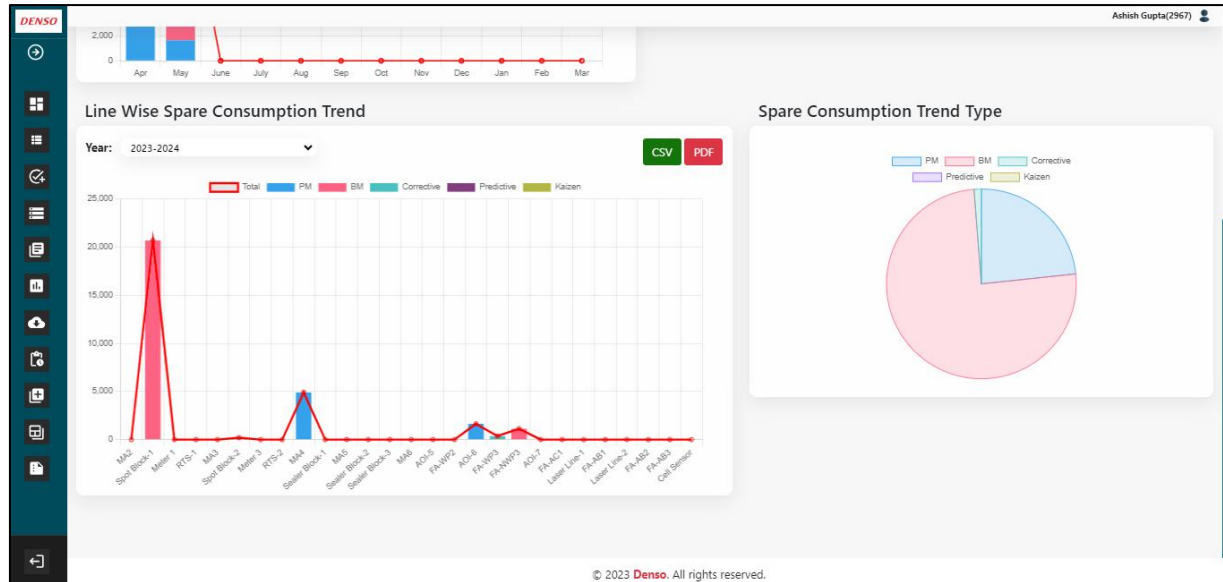
- Moreover, users can set monthly targets for line productivity in terms of production hours, breakdown (BD) hours, Mean Time To Repair (MTTR), Mean Time Between Failures (MTBF), and breakdown percentage (BD%) to track performance effectively.
- These targets will be utilized in all reports requiring comparisons between set targets and actual values for performance analysis.
- Note:** Note: If the user fails to add targets for any line, the accuracy of the report data will be compromised.

3.2.17 Spare Reports

- This dashboard contain four charts and each chart it's own different data analysis.



- Monthly spare consumption trend :
 - This indicates first overall spare cost of the section with it's different type (PM, BM, CM, Kaizen) total cost.
 - Also user will see the data line wise total cost, if user need to see again overall cost as section wise then click on the total button.
- Top 20 machine (Spare part consumption) :
 - On this table display top 20 machine which take higher cost of the spare.
 - Also, user will able to see past year data of the top 20 machine based on selection of the year.



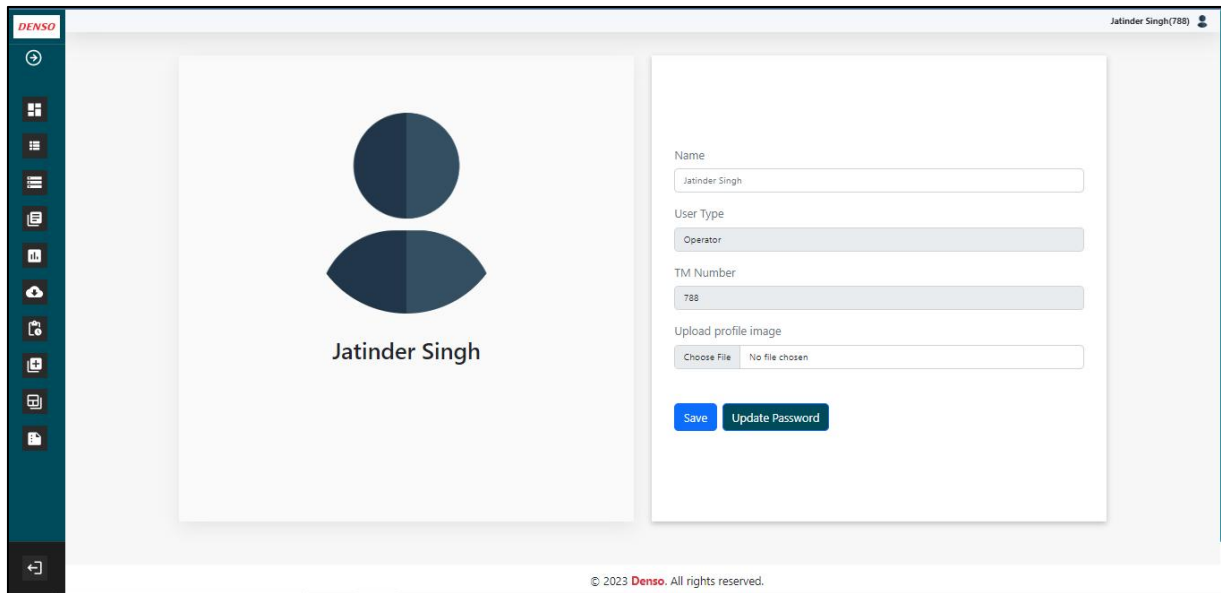
- Line wise spare consumption trend :
 - Line wise spare consumption as name suggest it display data as every line of the section with every types of spare usage total.
 - Also, user will able to see past year data of every line of the section based on selection of the year.
- Spare consumption trend type:
 - This charts indicate total cost as spare consumption type (PM, BM, CM , Kaizen).

3.2.18 Spare history

DENSO Integrated Maintenance Operation System												
Spare Usage History												
Select Cell 2024-2025 Month RESET												
Sr. No.	Date	Line	Machine	Machine No.	Category	Part Name	Part No.	Used By	Cost	Abnormality	SparePart	Action
									790027.45			
1	21/4/2024 - 9:53 AM	MA2	CM212-2	EETP-015	PM	O Ring	KXF0CT4AA00	Indrajit Behera	4000	Yes	Yes	
2	18/4/2024 - 5:28 am	MA2	YS Unloader	EETP-029	PM	ELEMENT	AF20P-060S	Dalip Khajuria	187	Yes	Yes	
3	21/4/2024 - 11:48 am	MA2	CM212-3	EETP-016	PM	PHOTO SENSOR W/CONNECTOR	N610017708AA	Indrajit Behera	3400	Yes	Yes	
4	21/4/2024 - 12:17 PM	Spot Block-2	Air Blowing M/C06	EEAT-047	PM	Element	AF60P-060S	Ujjawal	780	Yes	Yes	
5	9/4/2024 - 12:00 AM	MA2	CM212-1	EETP-014	Corrective	Element	AFM40P-060AS	Jatinder Singh	802			
6	9/4/2024 - 12:00 AM	MA2	CM212-2	EETP-015	Corrective	ELEMENT	AFM40P-060AS	Jatinder Singh	802			
7	21/4/2024 - 12:00 AM	MA2	CM212-2	EETP-015	Corrective	PHOTO SENSOR W/CONNECTOR	N610017708AA	Indrajit Behera	3566			

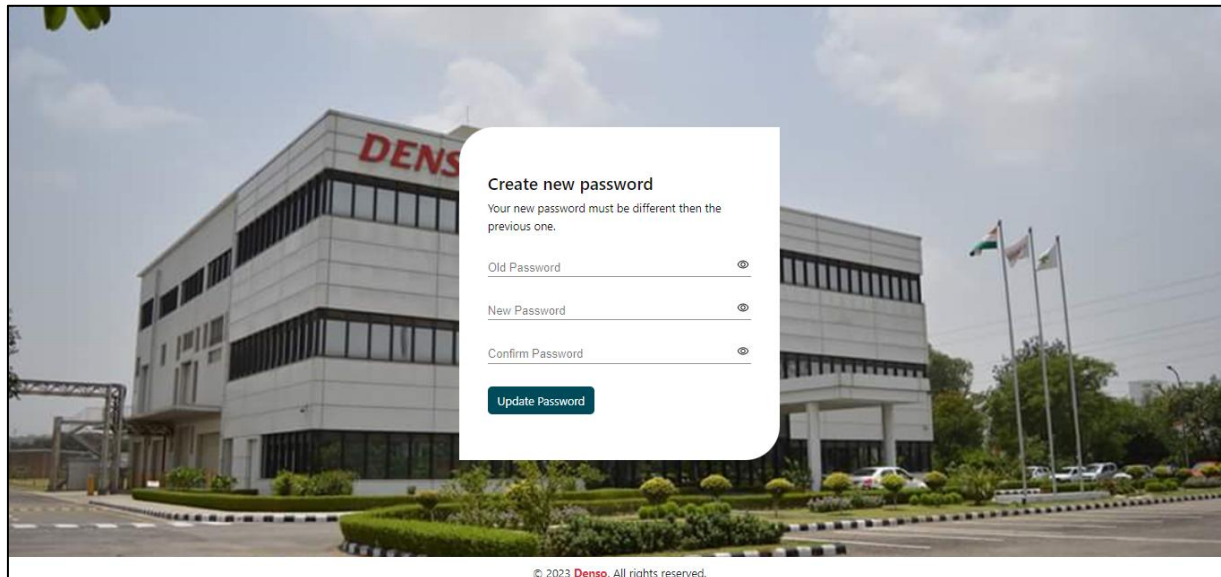
- Display the complete history of used spare parts, including preventive maintenance (PM), breakdown maintenance (BM), and corrective maintenance (CM), along with operator data entered by users.
- Additionally, display the total cost of spare parts at the top, providing an overview of expenses incurred.

3.2.19 Profile



The screenshot shows a web application interface for a user profile. On the left is a dark teal sidebar with various icons. The main content area has a light gray background. On the left side of this area is a large circular profile picture placeholder with the name 'Jatinder Singh' below it. To the right is a white form with the following fields: 'Name' (containing 'Jatinder Singh'), 'User Type' (a dropdown menu showing 'Operator'), 'TM Number' (containing '788'), and 'Upload profile image' (with a 'Choose File' button and 'No file chosen' text). At the bottom of the form are two buttons: 'Save' and 'Update Password'. The top right corner of the page shows the user's name 'Jatinder Singh(788)' and a user icon. The bottom center of the page has a copyright notice: '© 2023 Denso. All rights reserved.'

- User will be able to update their name and profile image on this dashboard.



The screenshot shows a 'Create new password' dialog box overlaid on a background image of a Denso building. The dialog box is white with a rounded bottom. It contains the title 'Create new password', a note 'Your new password must be different than the previous one.', and three password input fields: 'Old Password', 'New Password', and 'Confirm Password'. Each field has a toggle icon on the right. At the bottom of the dialog is an 'Update Password' button. The background image shows a modern white building with 'DENSO' in red letters on the roof, surrounded by greenery and flags. The bottom center of the page has a copyright notice: '© 2023 Denso. All rights reserved.'

- Also, user will be able to update their password. Need to add first old password then new.

3.2.20 Need to Know

- Not every user able to access every dashboard mention in user manual.
- Every approval on BM should be done one after one and also receive the mail regarding the every approval.
- All necessary targets must be included by the user; otherwise, the data will be inaccurate..
- Users have the capability to export reports as PowerPoint presentations (.pptx).
- When filtering data based on date selection in the report, users need to click the "Go" button each time they make a selection. Additionally, if they change the top filter, they must click the "Go" button again for the date filter.

4 APPLICATION DEPLOYMENT INSTRUCTIONS

Kindly follow the below steps to run the application –

- Double-click on batchFileDENSO_PM.bat file, the project will run.
- Note : If user need to stop the running project, press ctrl+c on opened command prompt.
- If user need to change server configuration then first take backup to the any particular folder and open command prompt and type below commands:

Backup or restore step:

1) Take dump of the database (Backup / Export):

- a) `mongodump --host=localhost --port=27017 --db=DENSO-PM-BM`

2) Restore database from the dump:

- a) Redirect up-to dump folder and then open command prompt and write this command:
 - i. `mongorestore dump`

Setup (Installations) required -

Node version: 16.13.2

NPM version: 8.1.2

MongoDB version: 6.0.15

MongoDB Database Tool: 100.9.4

MongoDB Shell: 2.2.5